
The Physiology of Muscle

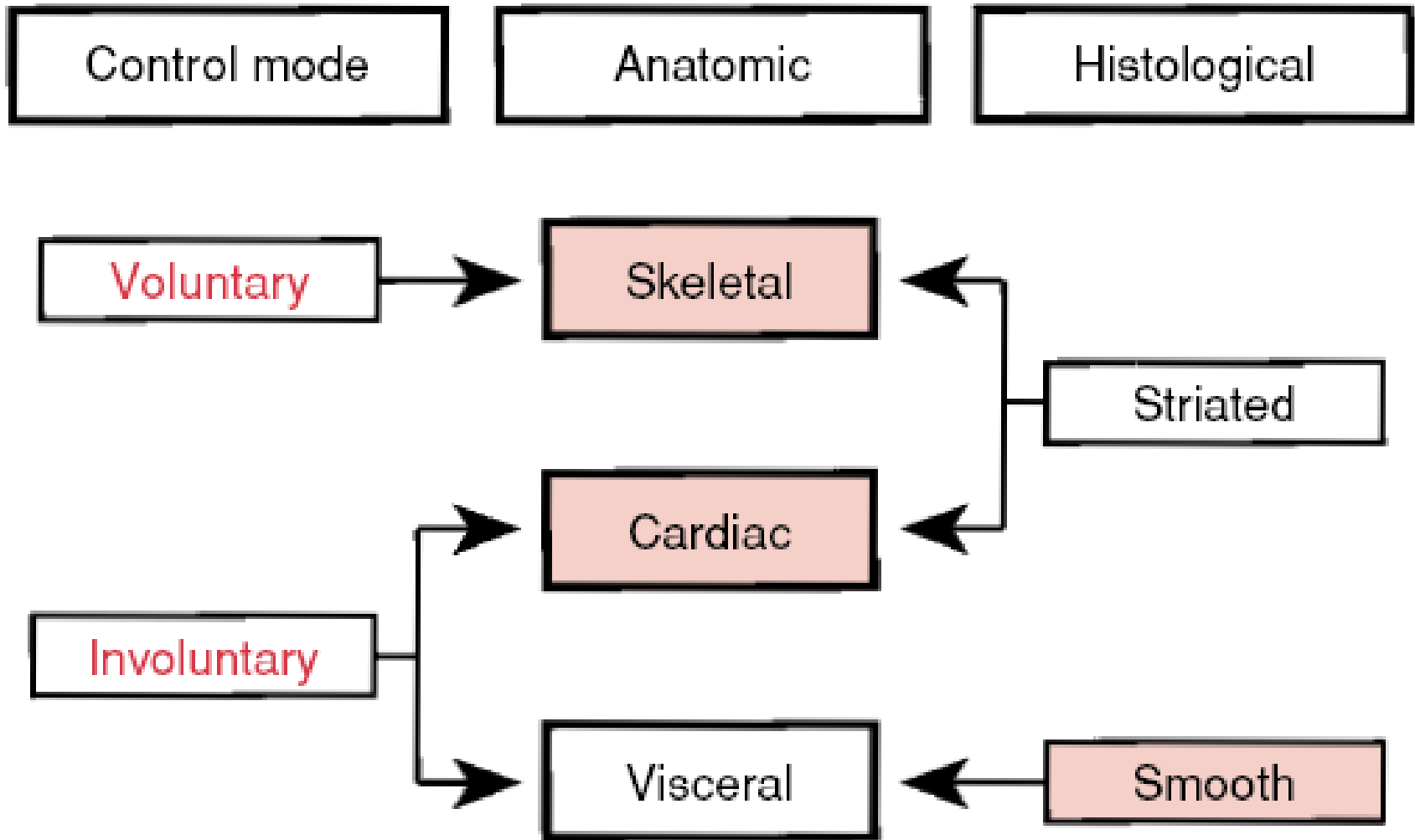
Characteristics

- **Excitability**
 - Ability to generate electrical impulse that travels along the sarcolemma in response to a stimulus (NTs, hormones, stretch...)
- **Contractility**
 - Ability to shorten when adequately stimulated (*defining property of muscle tissue*)
- **Extensibility** → ability to stretch (extend)
- **Elasticity** → ability to recoil and resume original length after being stretched or contracted

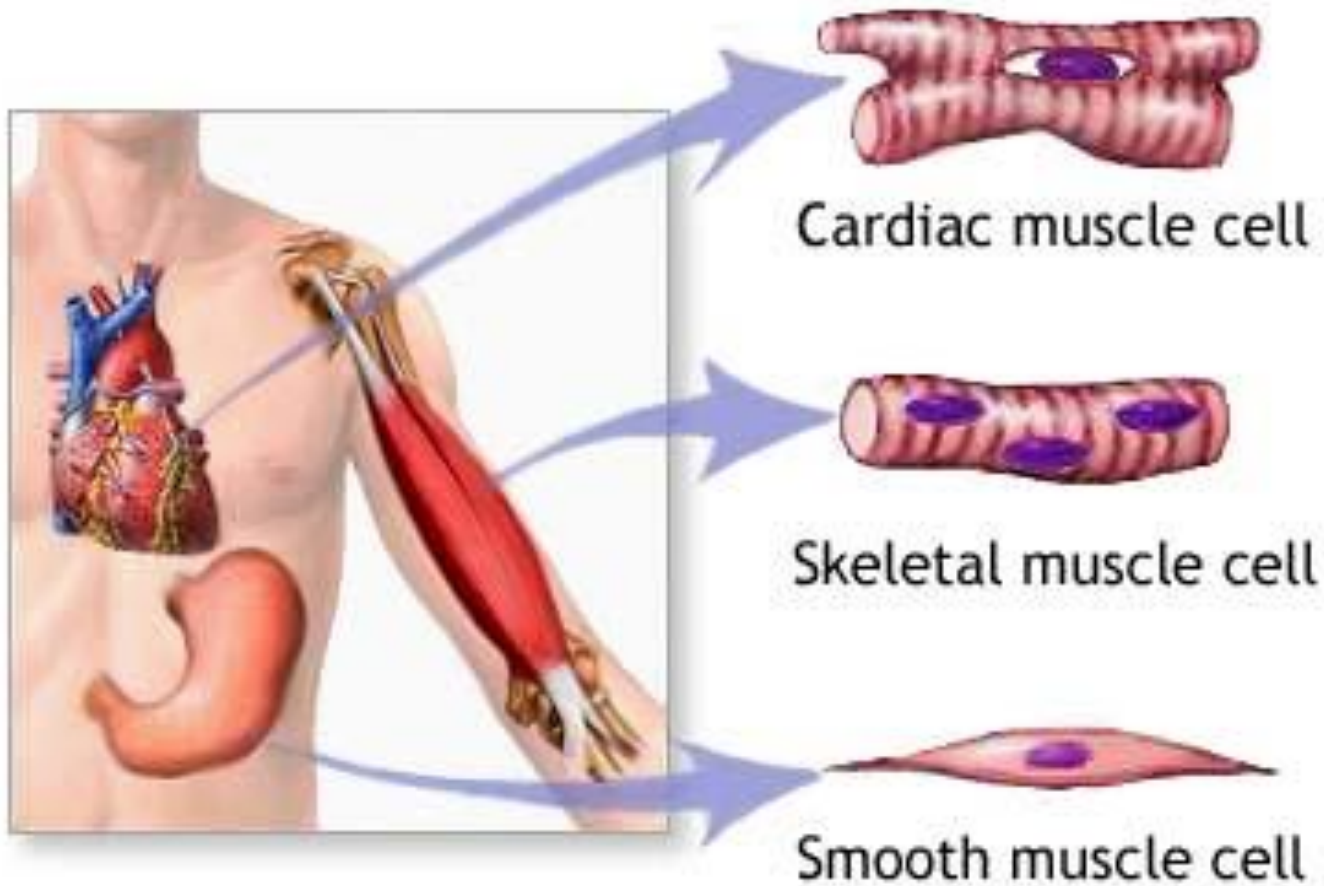
Function

- Movement
- Posture & Stability
- Thermogenesis

Classification



Classification

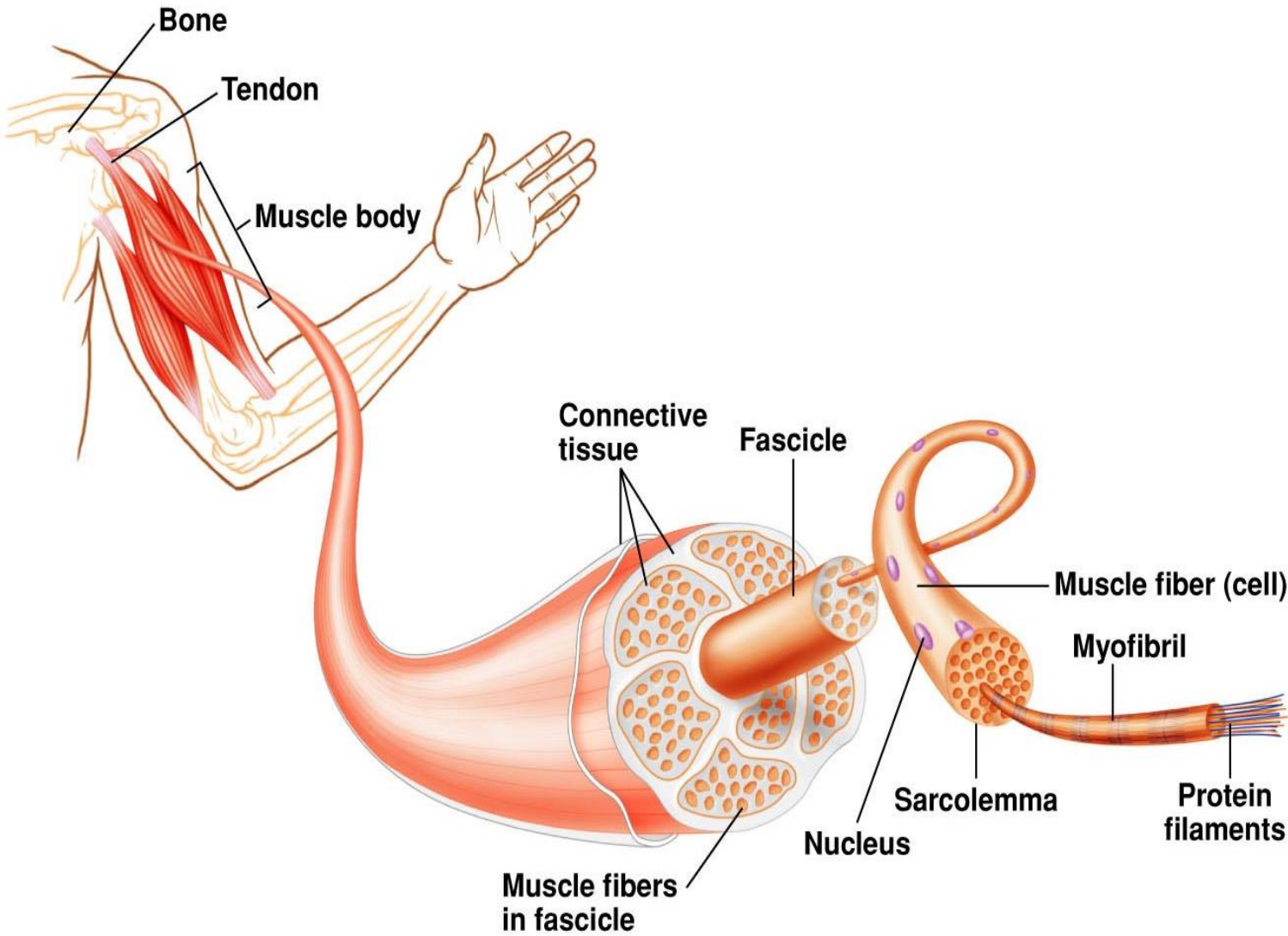


The Skeletal Muscle

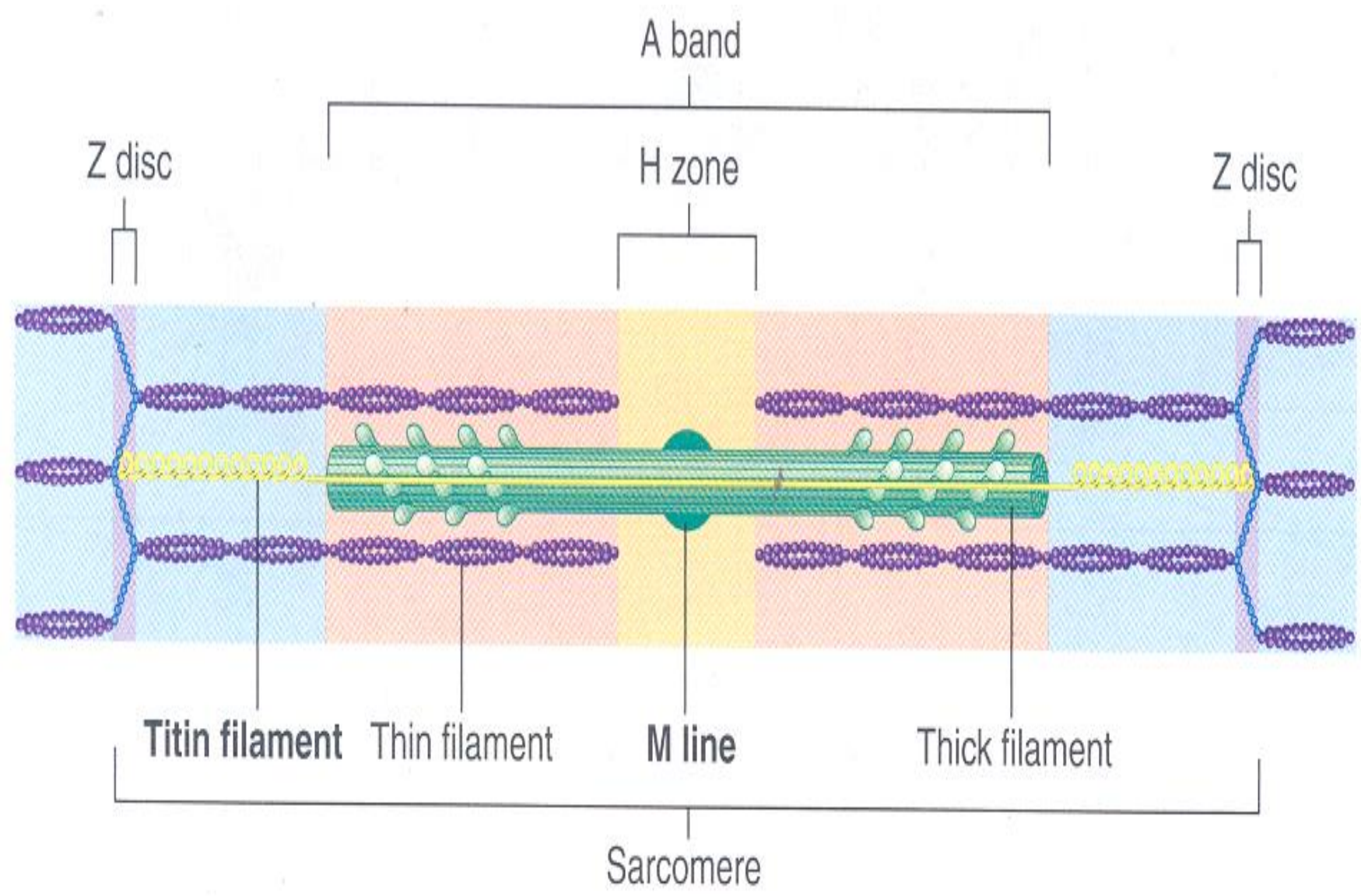
Skeletal Muscle

- **Characteristics:**
 - Dominant tissue of the skeletal system
 - Have well developed striations
 - Interdigitating dark and light bands on the sarcolemma
 - Voluntary muscle tissue (CNS, Somatic neurons)
 - Contract only in response to a stimuli:
 - No pacemaker cells + No syncytial bridges between cells
- **Function:**
 - Maintenance of posture and movement, speech, respiration...

Structural arrangement

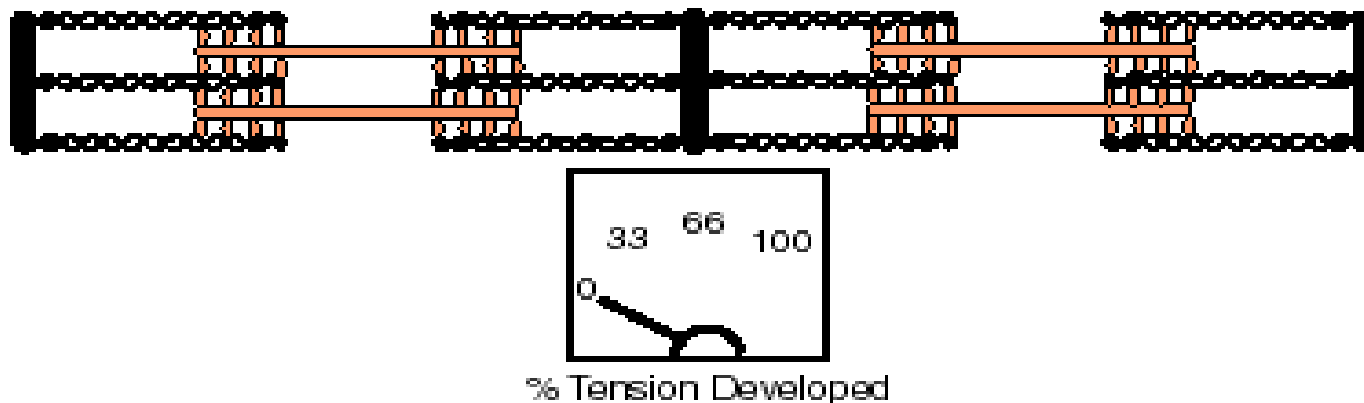


Geometrical orientation of the contractile elements



Molecular Geometry

- During muscular contraction:
 - There is **shortening of the sarcomere**
 - Length of I-band decreases and A-band remains constant
 - H-zone disappears and Z-line gets considerably darker
 - There is **no change** in the length of either thick or thin filaments



Molecular Basis of Contraction

- Thick Filament:
 - Myosin (an actin-binding protein)
 - Composition:
 - 2 heavy chains → form 1 long tail → form the core of the thick filament
 - 4 light chains → form 2 heads:
 - Myosin head (cross-bridge) → Actin-binding site
 - ATP-binding site (ATPase) → Hydrolyzes ATP

Molecular Basis of Contraction

- Thin Filaments (Actins):

1. Actin:

- Globular-Actin (G-actin)
- Fibrillar-Actin (F-actin, a polymer of G-actins)

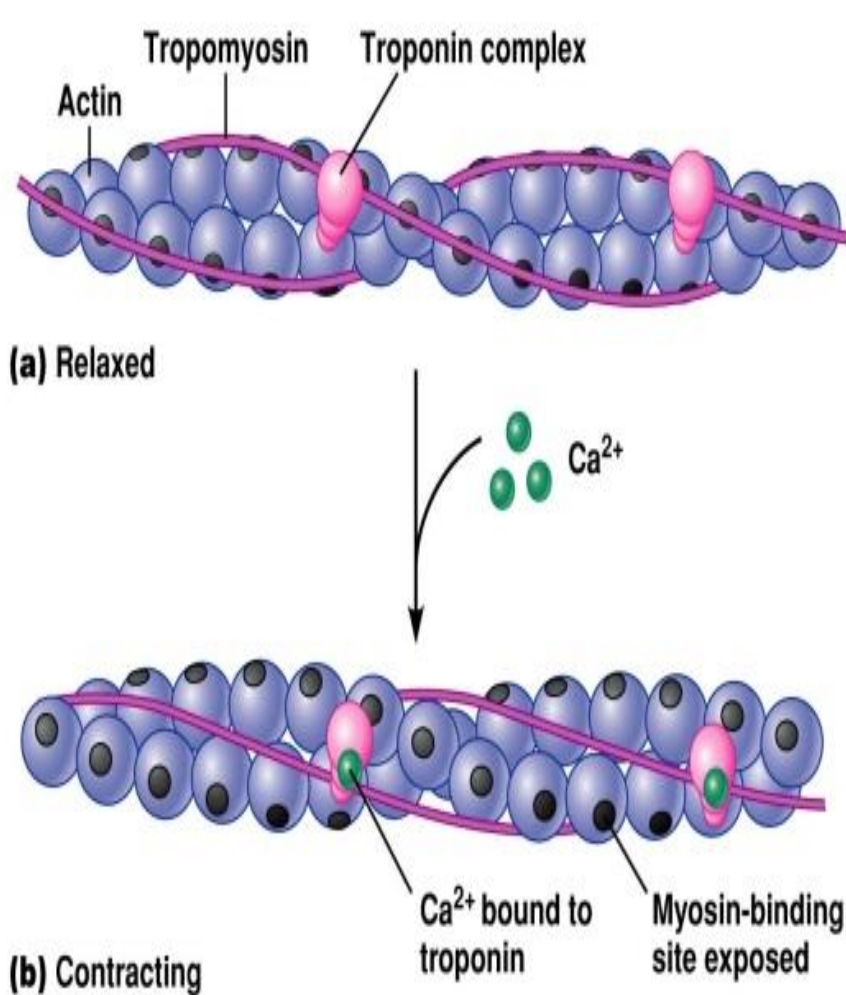
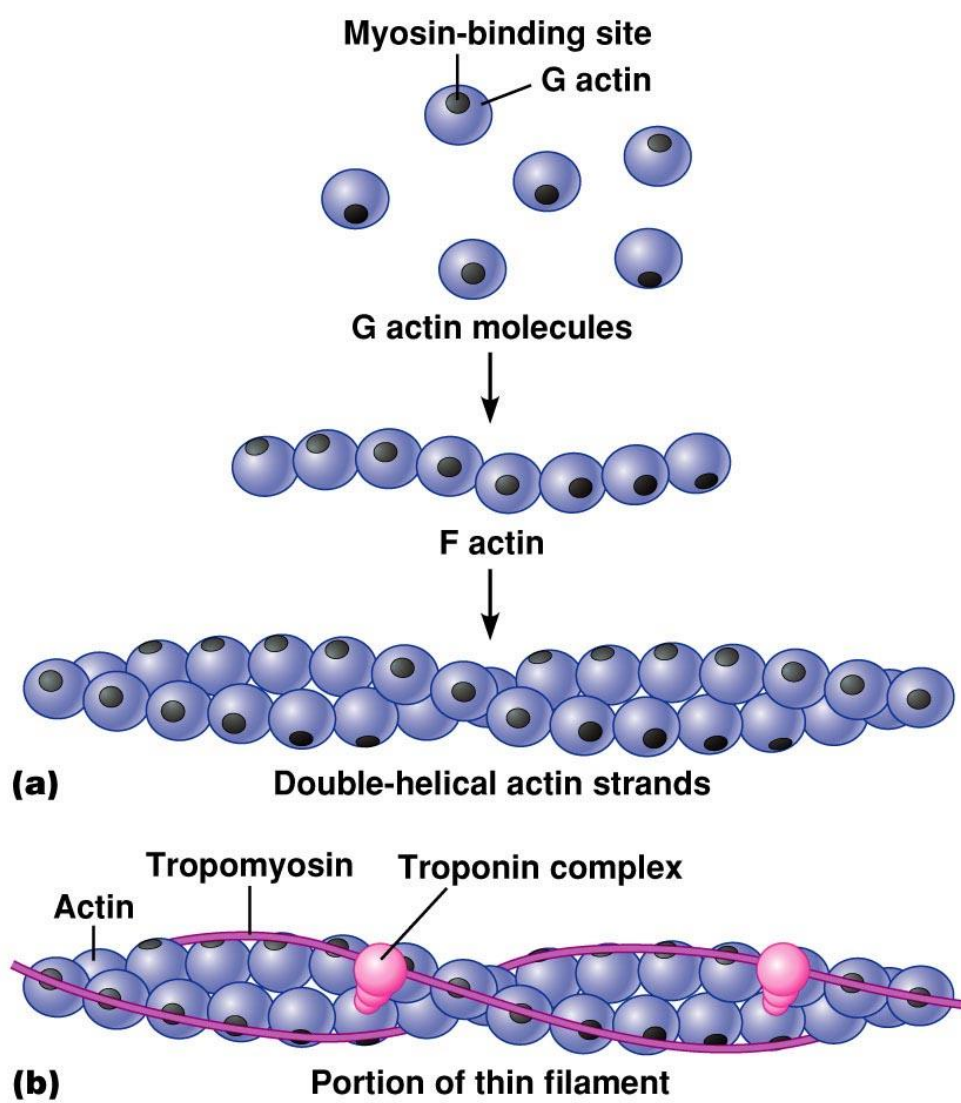
2. Troponins

- Small globular units located at intervals along the tropomyosin molecules
- **Troponin T** → binds other troponin components to tropomyosin
- **Troponin I** → inhibits the interaction of myosin with actin
- **Troponin C** → binds with Ca^{2+} and initiates contraction

Molecular Basis of Contraction

3. Tropomyosin:

- A rod-shaped non-globular proteins stretched along each strand of thin filament
- 1 tropomyosin molecule covers 7 actin monomers in an actin filament
- It blocks the binding sites of myosin on actin



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Neuromuscular Transmission

Skeletal Muscle Innervation

- Upper motor neurons (UMNs)
 - Neurons in the brain that influence control of skeletal muscles
 - Act by facilitating or inhibiting (through interneurons) the activity of LMNs
- Lower motor neurons (LMNs)
 - Neurons whose axons innervate skeletal muscles
 - Two types of LMNs innervate skeletal muscles:
 1. α -motor neurons
 2. γ -motor neurons

Skeletal Muscle Innervation

1. α -motor neurons

- Innervate the extrafusal (ordinary) muscle fibers
- Faster conducting (70-120m/s)
- Activation of α -motor neurons **causes muscle contraction**

2. γ -motor neurons

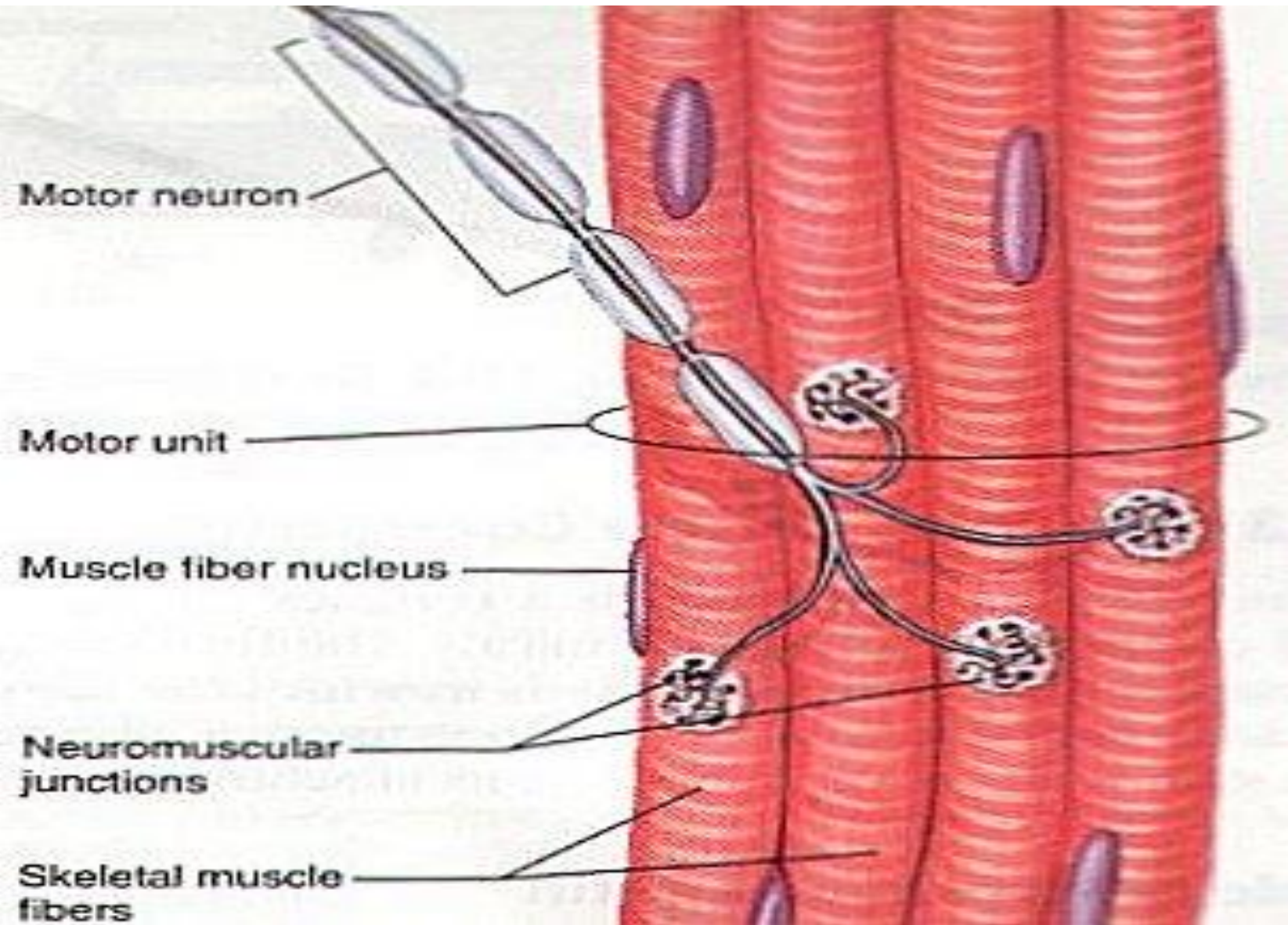
- Innervate the contractile polar regions of intrafusal fibers
- Slower conducting (15-40m/s)
- Control **sensitivity** of muscle receptors to stretch
 - Activation of γ -motor neurons increases **firing rate of sensory** endings

Skeletal Muscle Innervation

- Activity of LMNs is influenced by:
 - Sensory feedback from the muscles and tendons
 - Facilitatory and inhibitory effects from UMNs that contribute axons to descending motor tracts:
 - Pyramidal tracts
 - Extrapyramidal tracts
- LMNs are the **final common pathway** by which sensory stimuli and higher brain centers exert control over skeletal movements

Skeletal Muscle Innervation

- **Motor unit:**
 - Motor neuron and all the muscle fibers that it innervates
 - In general each muscle is served by one nerve – a bundle of axons carrying signals from the spinal cord to the muscle
 - Within the muscle, each axon will go its own way and branch into multiple small extensions and ends in the synaptic end bulb
 - **Neuromuscular junction** → synapse between a neuron and a muscle



Skeletal Muscle Innervation

- **Motor unit:**
 - The number of muscle fibers innervated by a single motor neuron varies from a few to thousands
 - The fewer the number of fibers per neuron (**smaller motor unit**) → the finer and more delicate the movements (*more neural control*)
 - **Extraocular muscles** typically have small motor units while the large **postural muscles** have large motor units

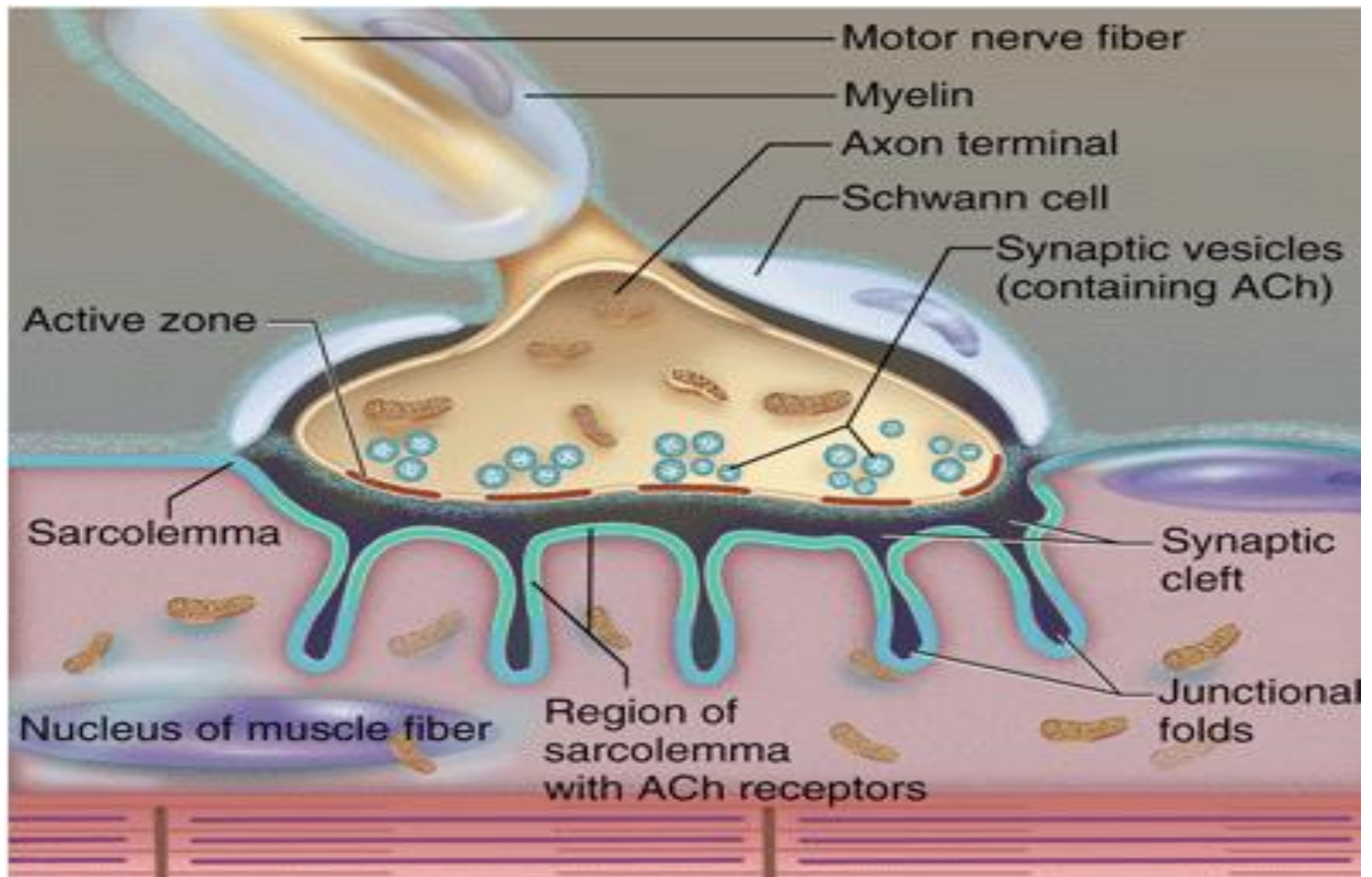
Neuromuscular Transmission

- A synaptic transmission between α -motor neuron and a skeletal muscle fiber

1. Presynaptic end

- Neurotransmitter $\rightarrow$ Acetylcholine (Ach)
- Ach is synthesized in the soma of α -motor neurons and transported to the presynaptic terminal
- Ach is contained in synaptic vesicles and concentrated at the active zone (presynaptic terminal)

Neuromuscular Transmission

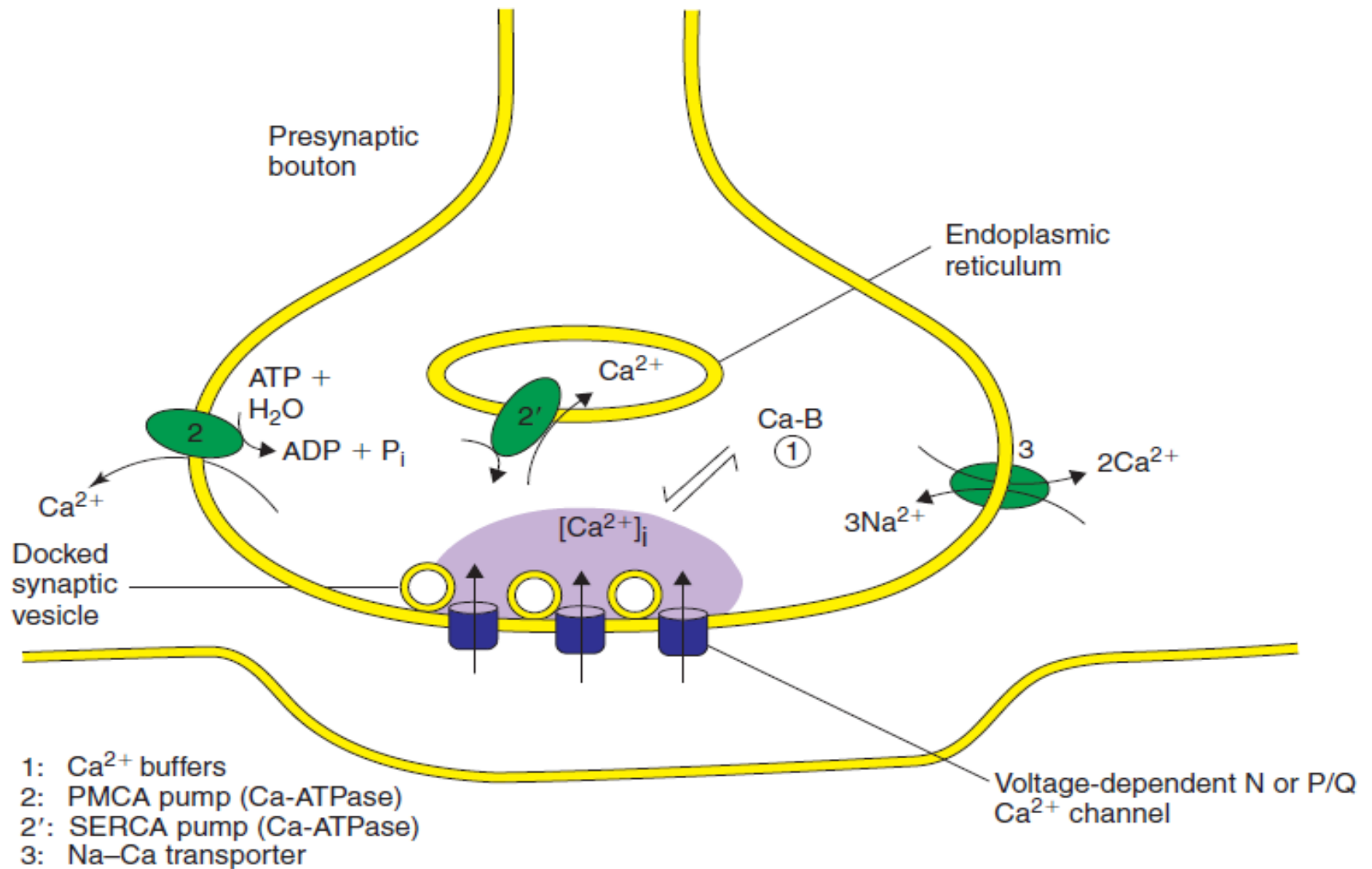


NT Release

- Transmitter release is regulated by **depolarization** of the presynaptic terminal and triggered by **calcium influx**
- **Duration of the AP** regulates the amount of Ca^{2+} that flows into the presynaptic terminal + the amount of transmitter release
- The release of transmitter from a vesicle requires the binding of 4Ca^{2+} to a calcium-sensing synaptic vesicle protein
- The **amount of transmitter release** is a function of the concentration of Ca^{2+} in the presynaptic terminal
 - The greater the Ca^{2+} influx into the terminal, the larger the number of transmitter quanta released

NT Release

- Ca^{2+} clearance mechanisms in a presynaptic terminal;
 - Ca^{2+} buffering by calcium-binding proteins
 - Ca^{2+} Pump (Ca^{2+} -ATPase)
 - SR Ca^{2+} -ATPase (SERCA Pump)
 - Na^{+} - Ca^{2+} Exchanger (3Na^{+} in, 2Ca^{2+} out)



Ca^{2+} clearance mechanisms in a presynaptic terminal

Neuromuscular Transmission

- Events at presynaptic end

AP in presynaptic α -motor neuron terminals



Depolarization of plasma membrane of the presynaptic α -motor neuron axon terminals



Opening of Ca^{2+} channels at the active zones $\rightarrow$ $\uparrow$ Ca^{2+} permeability and entry of Ca^{2+} into α -motor neuron axon terminals



Release of Ach from the synaptic vesicles into the synaptic cleft

Neuromuscular Transmission

2. Synaptic cleft

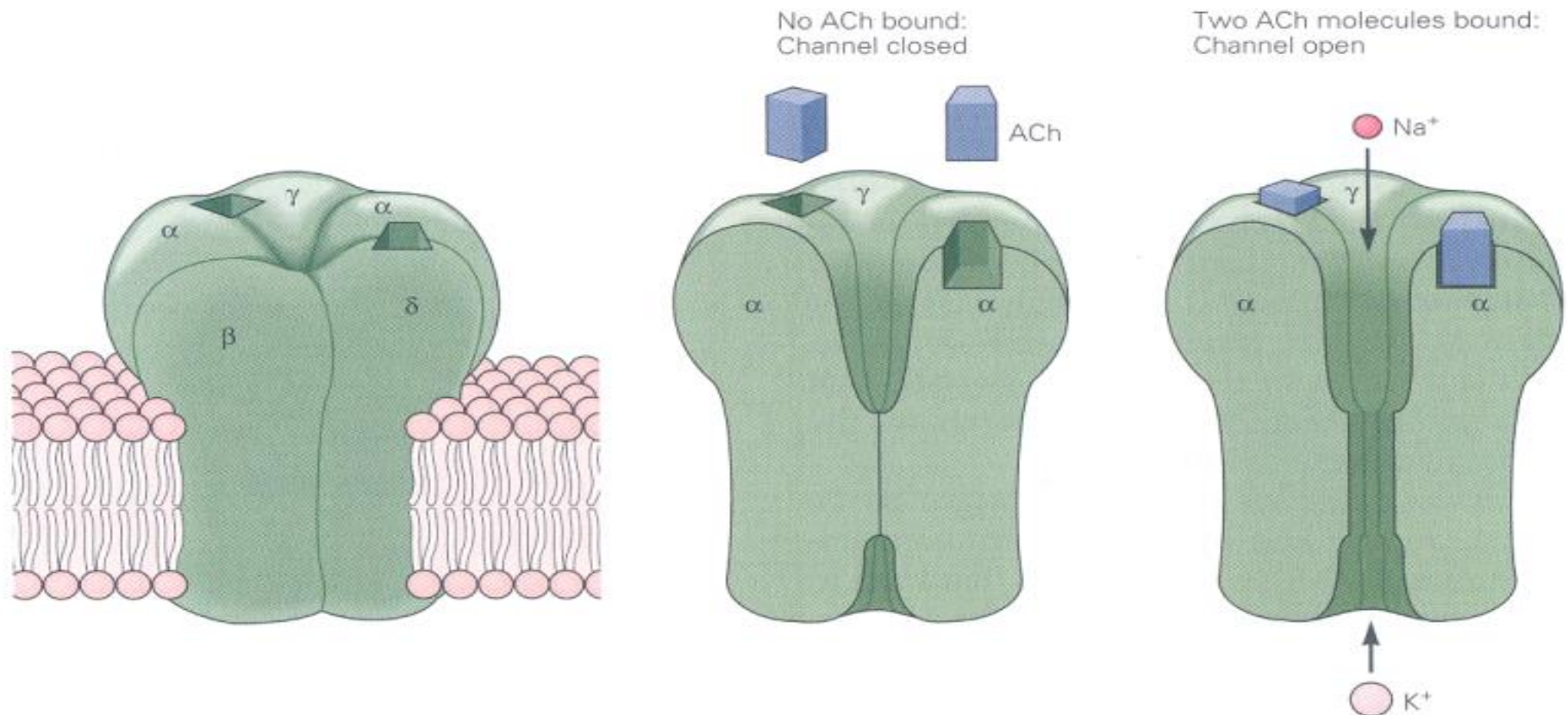
- Amorphous network of connective tissue, basal lamina
- Inactivating enzymatic system → *Acetylcholinesterase (AChE)*

3. Postsynaptic membrane

- Junctional folds → membrane invaginations located opposite to the active zones → ↑ *Surface Area*
- Acetylcholine receptors → *nAChR*

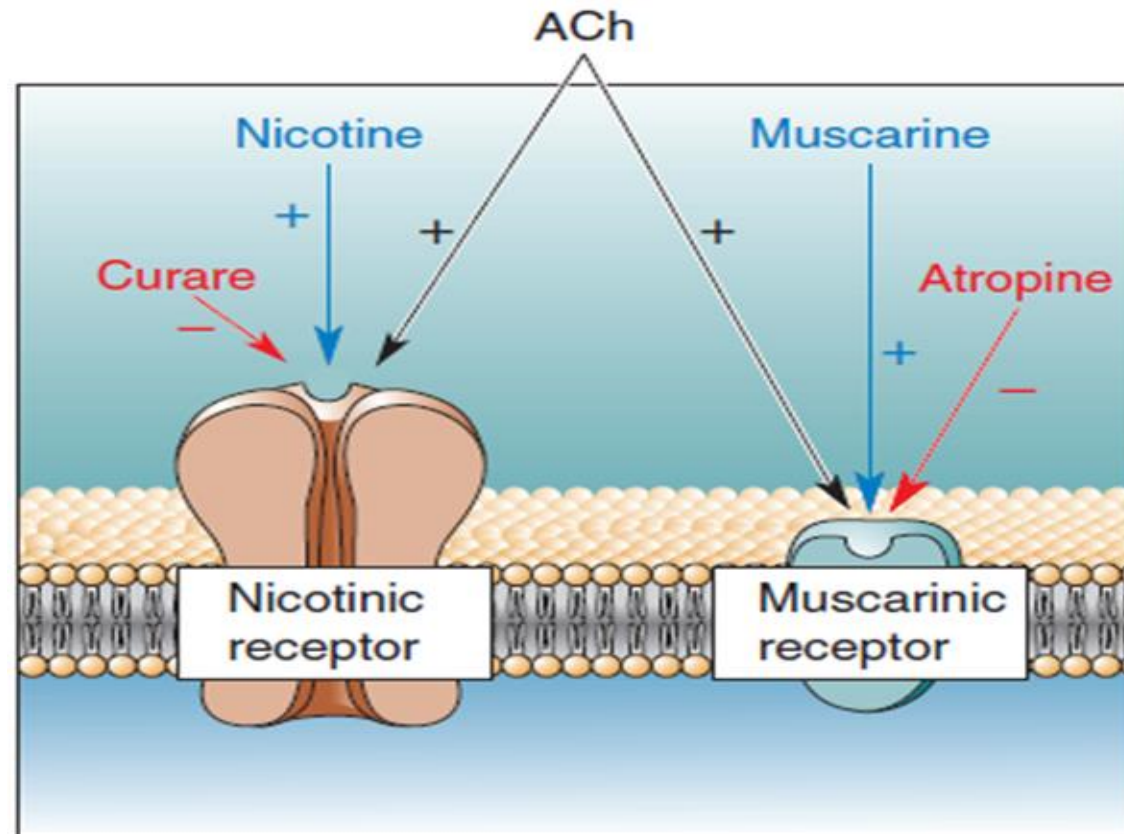
Neuromuscular Transmission

- Nicotinic ACh receptor:
 - Have 5 subunits $\rightarrow 2\alpha$ (ACh binding site) + $\beta + \gamma + \delta$



Neuromuscular Transmission

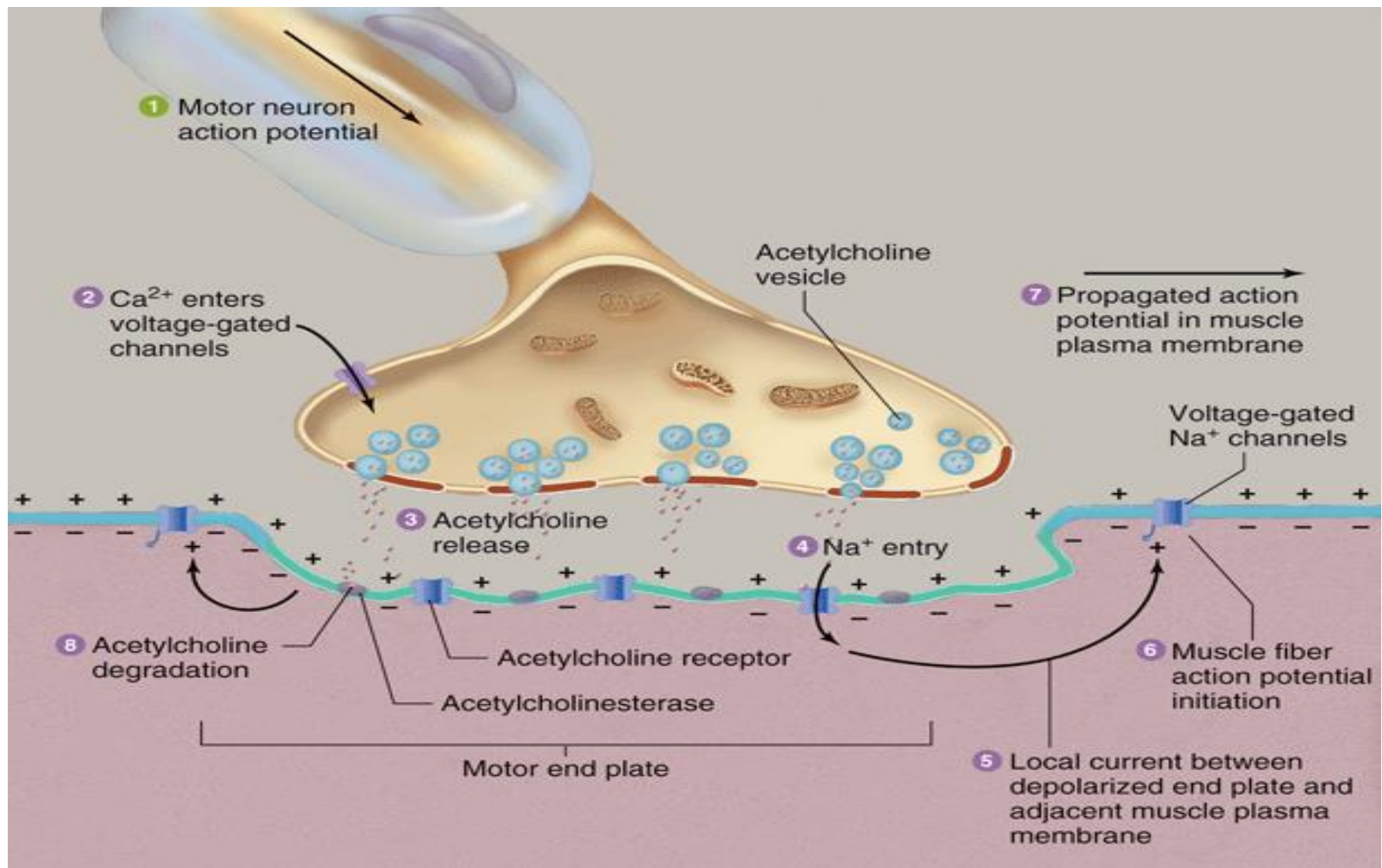
- Nicotinic ACh receptor:
 - Ionotropic receptor → functions both as a **receptor and an ion channel**
- Location:
 - NMJ
 - Autonomic ganglia
 - CNS



Neuromuscular Transmission

- Ach Effects:
 - Ach binds to the 2α subunits $\rightarrow$ conformational change $\rightarrow$ opening of the gate $\rightarrow$ permeability to Na^+ , K^+
 - Electrochemical gradient for $\text{Na}^+ >$ electrochemical gradient for $\text{K}^+ \rightarrow \text{Na}^+$ enters the cell $\rightarrow$ cell depolarizes $\rightarrow$ **EPP** $\rightarrow$ electrotonic conduction of current to the adjacent muscle membrane $\rightarrow$ activation of VG Na^+ channels $\rightarrow$ generation and propagation of AP along the sarcolemma $\rightarrow \rightarrow \rightarrow$

Neuromuscular Transmission



Neuromuscular Transmission

- Endplate Potential (EPP)
 - A graded response (magnitude of depolarization $\propto$ number of open Ach channels)
 - Transient ($\because$ Ach is hydrolyzed to form choline and acetate)
 - No VG Na^+ and K^+ channels on the endplate region ($\therefore$ no AP)
 - VG Na^+ , K^+ channels are located on the muscle membrane contiguous to the endplate
 - EPP depolarizes the contiguous membrane regions by electrotonic conduction to threshold, and an AP is generated

Neuromuscular Transmission

- Events at postjunctional end:

Diffusion of Ach to Postjunctional membrane → combination of Ach with n AchR on the motor endplate



Conformational change → opening of the gate → ↑ permeability of the postjunctional membrane to Na^+ and K^+



Transient change in the E_m → depolarization → EPP

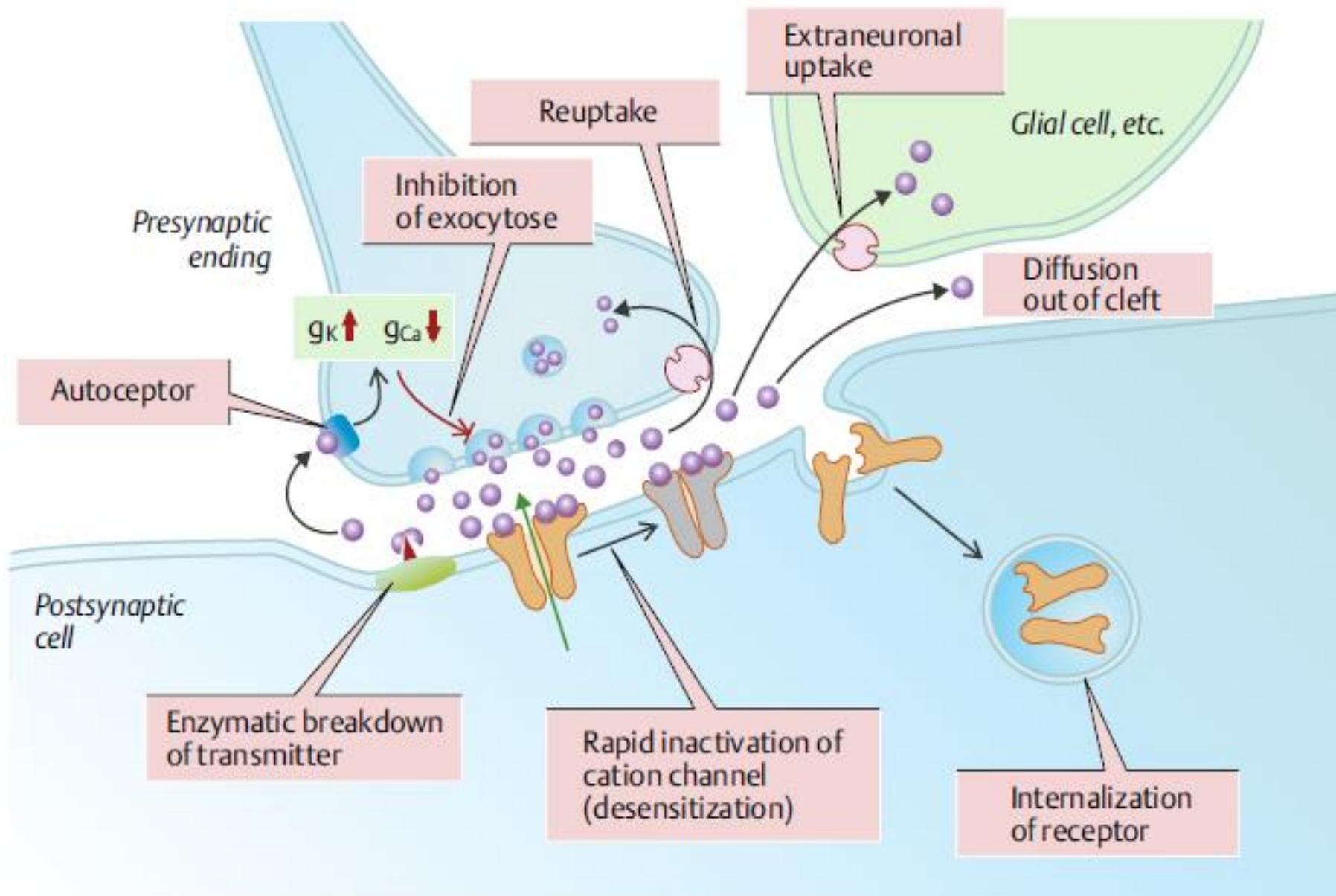


Depolarization of areas of muscle membrane adjacent to endplate and initiation of AP → muscle contraction

Neuromuscular Transmission

- **Fates of Ach:**
 - Enzymatic inactivation (AChE)
 - Re-uptake by presynaptic nerve terminal
 - Extra-neural uptake (glial cells)
 - Diffusion out of the synaptic cleft
- **NB:**
 - Removal of Ach is necessary to prevent it from causing multiple contractions + Prevents receptor desensitization

The Fate of Acetylcholine



Neuromuscular Transmission

Clinical Correlates

Myasthenia Gravis

- **Autoimmune** disease where antibodies attack **nicotinic Ach receptors** on the NMJ
- Results in progressive weakening of the skeletal muscles
- Characterized by easy fatigability of the skeletal muscle
- Abs against *nAChR* → ↓ *nAChR* on the post-junctional membrane → ↓EPP → weakness and inability to sustain muscle contraction
- **Treatment:**
 - Anticholinesterases: Neostigmine or physostigmine

Lambert-Eaton Syndrome

- An autoimmune disease associated with **muscle weakness**
- **Antibodies against VG Ca^{2+} channels** in the presynaptic nerve endings at NMJ $\rightarrow$ $\downarrow$ density of VG Ca^{2+} channels
- $\downarrow \text{Ca}^{2+}$ influx $\rightarrow$ $\downarrow$ Ach release $\rightarrow$ muscle weakness

Botulinum Toxin

- Produced by anaerobic **clostridium botulinum** bacteria
- A specific protease that **cleaves SNARE** proteins present in the vesicular and presynaptic membranes which are involved in exocytosis
- Botulism **causes flaccid paralysis** by blocking the release of Ach at the NMJ
- Botulinum toxin preferentially affects **motor neurons**

Tetanus Toxin

- Produced by **clostridium tetani**
- Preferentially taken up by **inhibitory spinal interneurons**
- It is a protease that **cleaves SNARE proteins** involved in exocytosis, resulting in the abolition of inhibitory NT release in spinal interneurons
- Damage to inhibitory spinal interneurons results in disinhibition of spinal motor neurons → **Hyper-excitation of skeletal muscles** → **Tetanic Contractions** → **Spastic paralysis**

- *Curare*
 - Blocks nicotinic Ach receptors at the NMJ
 - Causes flaccid paralysis
- *Nerve gas*
 - Irreversibly inhibits acetylcholinesterase
 - Causes spastic paralysis

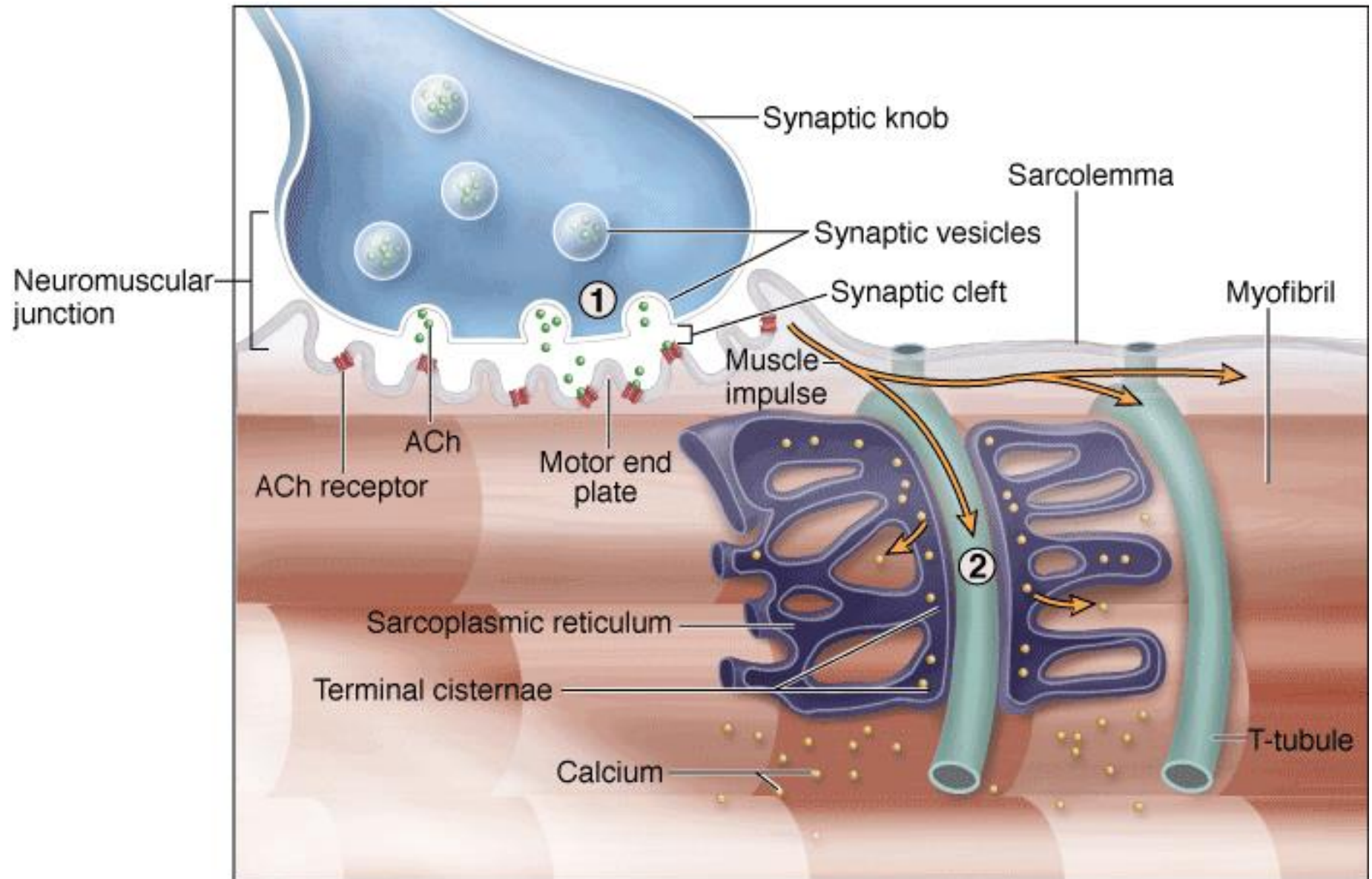
Excitation-Contraction Coupling

Excitation-Contraction Coupling

- ECC → the process of linking ΔE_m (AP) to muscle contraction
 - Coupling of electrical events with mechanical events
 - Electrical events precedes mechanical events (2ms, 100ms)
 - For the muscle to contract it need to be excited first
 - Once the muscle membrane at the NMJ is excited by α -motoneuron, the contiguous muscle membrane will be excited by electrotonic conduction
 - Then the AP generated will cause Ca^{2+} release from the SR and ultimately causes muscle contraction

Excitation

- Propagation of AP:
 1. Transverse Tubule (T-tube)
 - An invagination of the surface of the muscle membrane
 - It is found at the junction of **A-band and I-band**
 - It's one end is open to the extracellular space, but its other end is closed
 - **Function** → rapid transmission of the action potential from the cell membrane to all the fibers on the muscle



Excitation

- Propagation of AP:

2. Sarcoplasmic reticulum (SR)

- An internal tubular structure that runs between the myofibrils
- It is closed at both ends
- It is not continuous with the sarcolemma
- *Functions:*
 - Stores Ca^{2+} in the terminal cisternae (*1 calsequestrin* = 43 Ca^{2+})
 - *Calsequestrin* is sarcoplasmic protein that binds Ca^{2+}
 - Uptake and release of Ca^{2+}

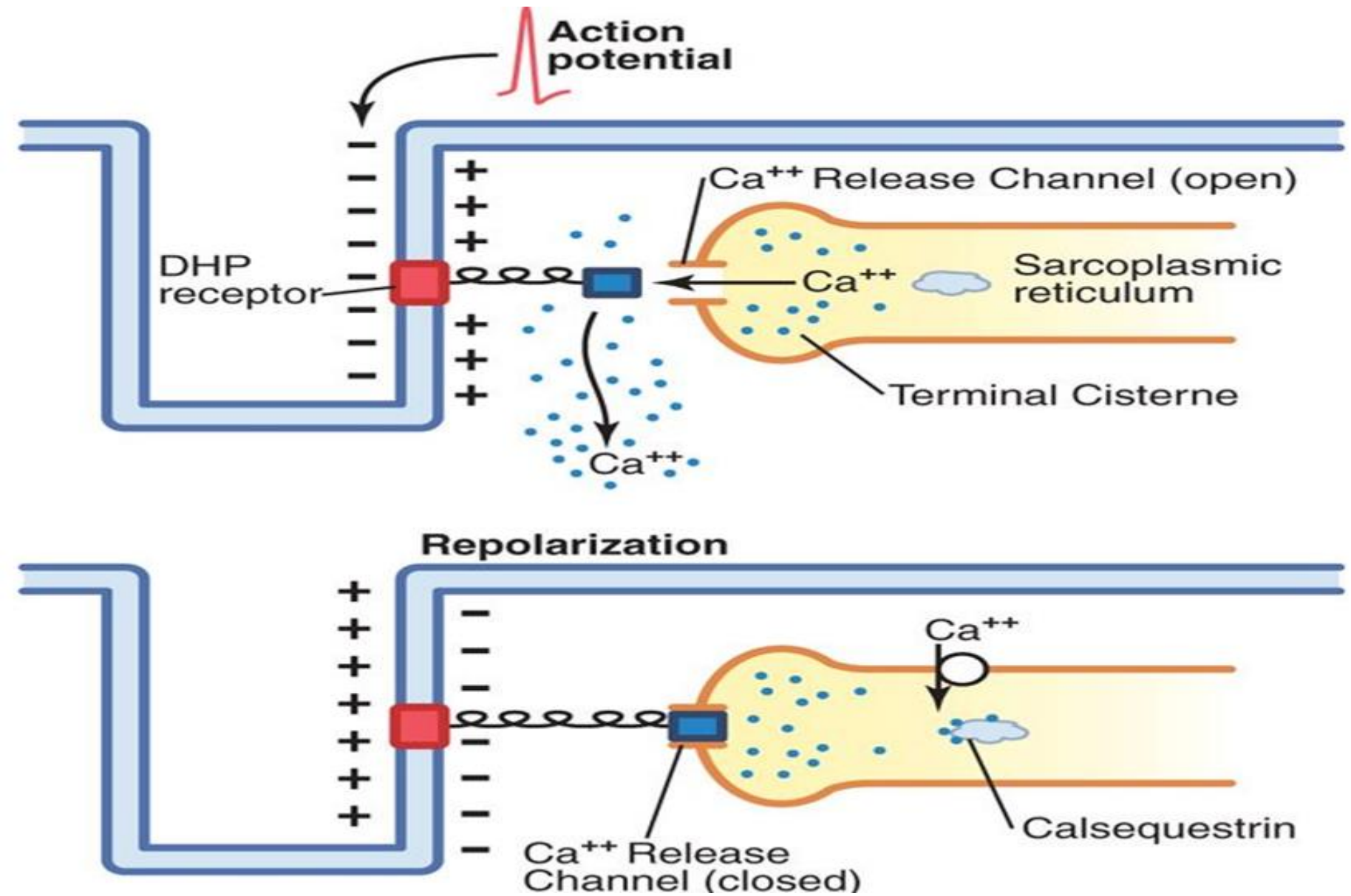
Excitation

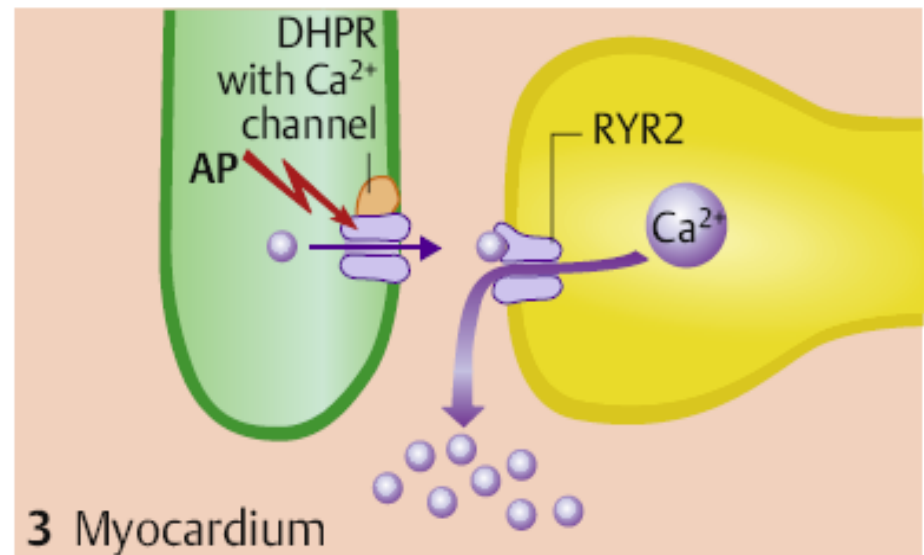
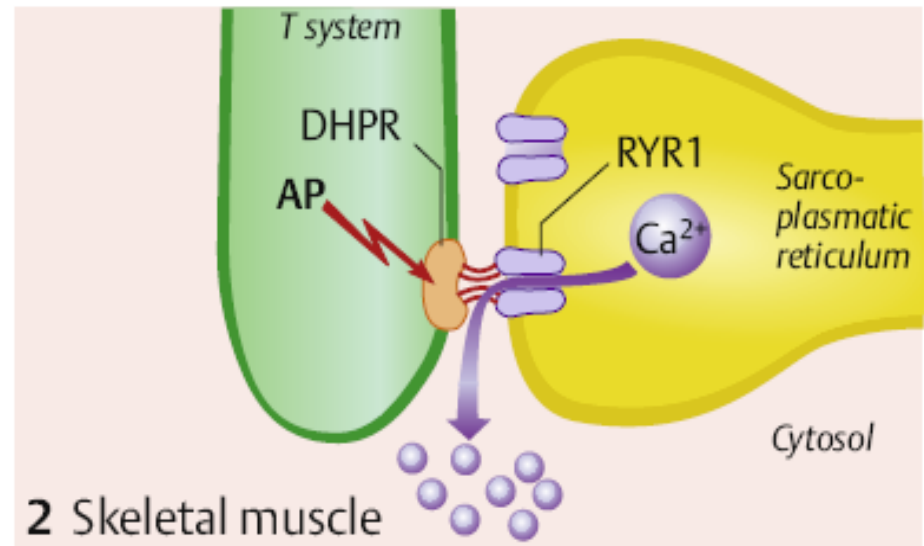
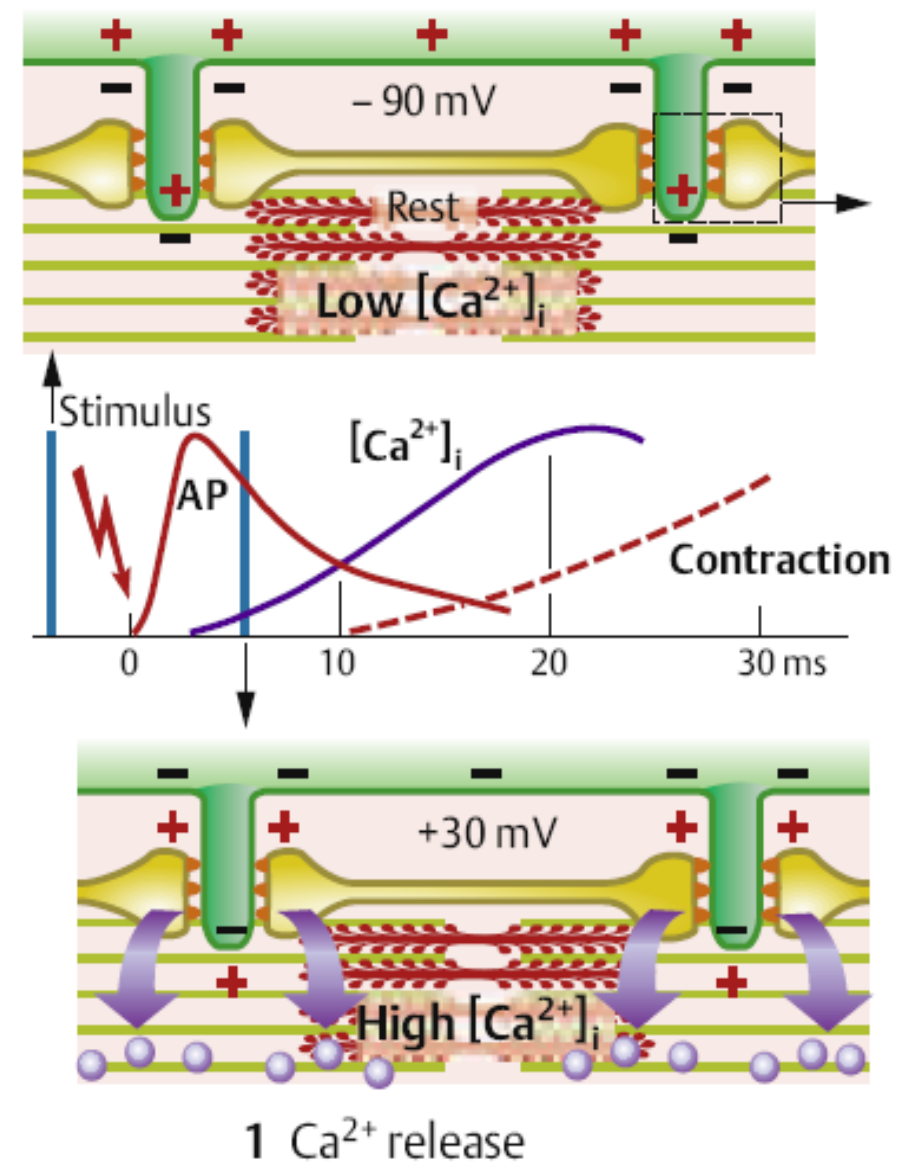
- Release of Ca^{2+} from the SR
1. Voltage - gated Ca^{2+} channel (DHPR)
 - Found in the transverse tubule
 - It is a pentamer ($\alpha_1, \alpha_2, \beta, \gamma, \delta$)
 - Voltage sensor which is responsible for triggering Ca^{2+} release from the SR and for EC coupling
 - Sensitive to a drug called Dihydropyridine which can block this voltage-gated channels

Excitation

2. Ca^{2+} - Release Channel (RyR)

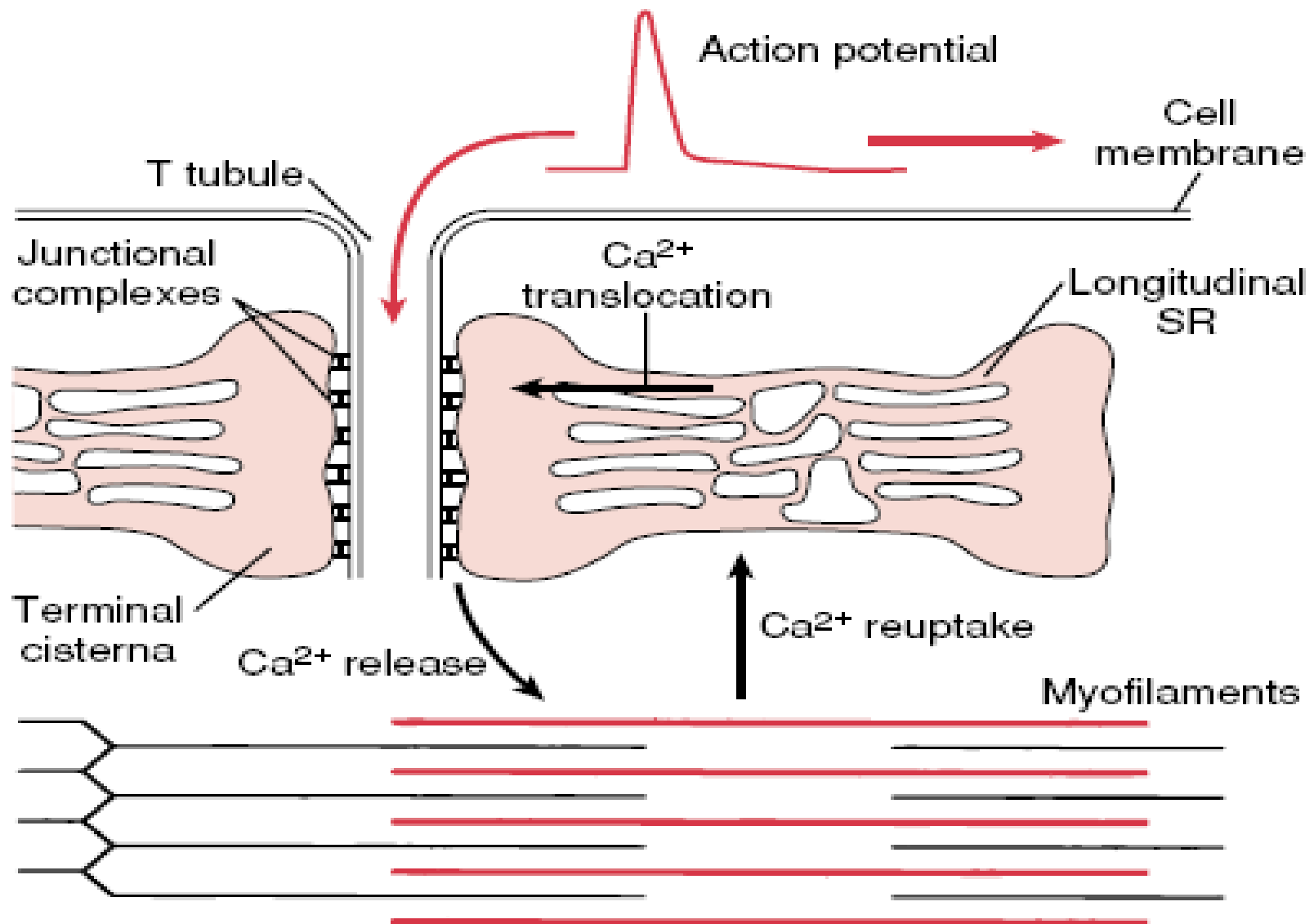
- Found in sarcoplasm reticulum membrane
- It is a tetramer
- Sensitive for a drug called Ryanodine
- Depolarization of the T-tubule $\rightarrow$ opening of DHPR - RyR Ca^{2+} channels (*Connected by a foot process whose length is 20nm*) $\rightarrow$ release of calcium from SR $\rightarrow$ Ca^{2+} flows out of the SR into the sarcoplasm $\rightarrow$ $\uparrow$ Sarcoplasmic $[\text{Ca}^{2+}]$





Activation of Muscle Proteins

- Resting state:
 - Interaction of thick and thin filaments is inhibited
 - Troponin I & Tropomyosin covers the sites where myosin heads bind to actin
- Activated States:
 - Influx of Ca^{2+} → binds to Troponin C (4Ca^{2+}) → conformational change in troponin → Tropomyosin moves aside → exposes the myosin-binding sites on actin → Myosin cross-bridge on the thick filament is exposed to actin filaments



Activation of Muscle Proteins

- Regulatory function of ATP
 - $\text{Actin} + \text{Myosin} + \text{ATP} + \text{Ca}^{2+} \longrightarrow \text{Contraction}$
 - $\text{Actin} + \text{Myosin} + \text{ATP} - \text{Ca}^{2+} \longrightarrow \text{Relaxation}$
- 3 ATP molecules are needed for contraction-relaxation:
 - For energizing the myosin cross-bridges
 - For dissociation of actin-myosin complex and initiation of relaxation
 - To pump out Ca^{2+} from the sarcoplasm to sequester it into the SR (SERCA)

Muscle Contraction

1. Muscle Twitch

- **Twitch** → contractile response of a muscle to a single supra/threshold stimulus
- A sub-threshold stimulus would not cause contraction because no AP would be produced
- **1 stimulation → 1 twitch**
 - The muscle quickly contracts and then relaxes
- A single muscle twitch produces only a very small force, Thus incapable of doing anything

Muscle Contraction

- Muscle twitch phases:

A. Latent phase

- Time between stimulus application and generation of tension
- Includes all the time required for excitation, E-C coupling, and activation of muscle proteins
- Myosin binding to actin active site

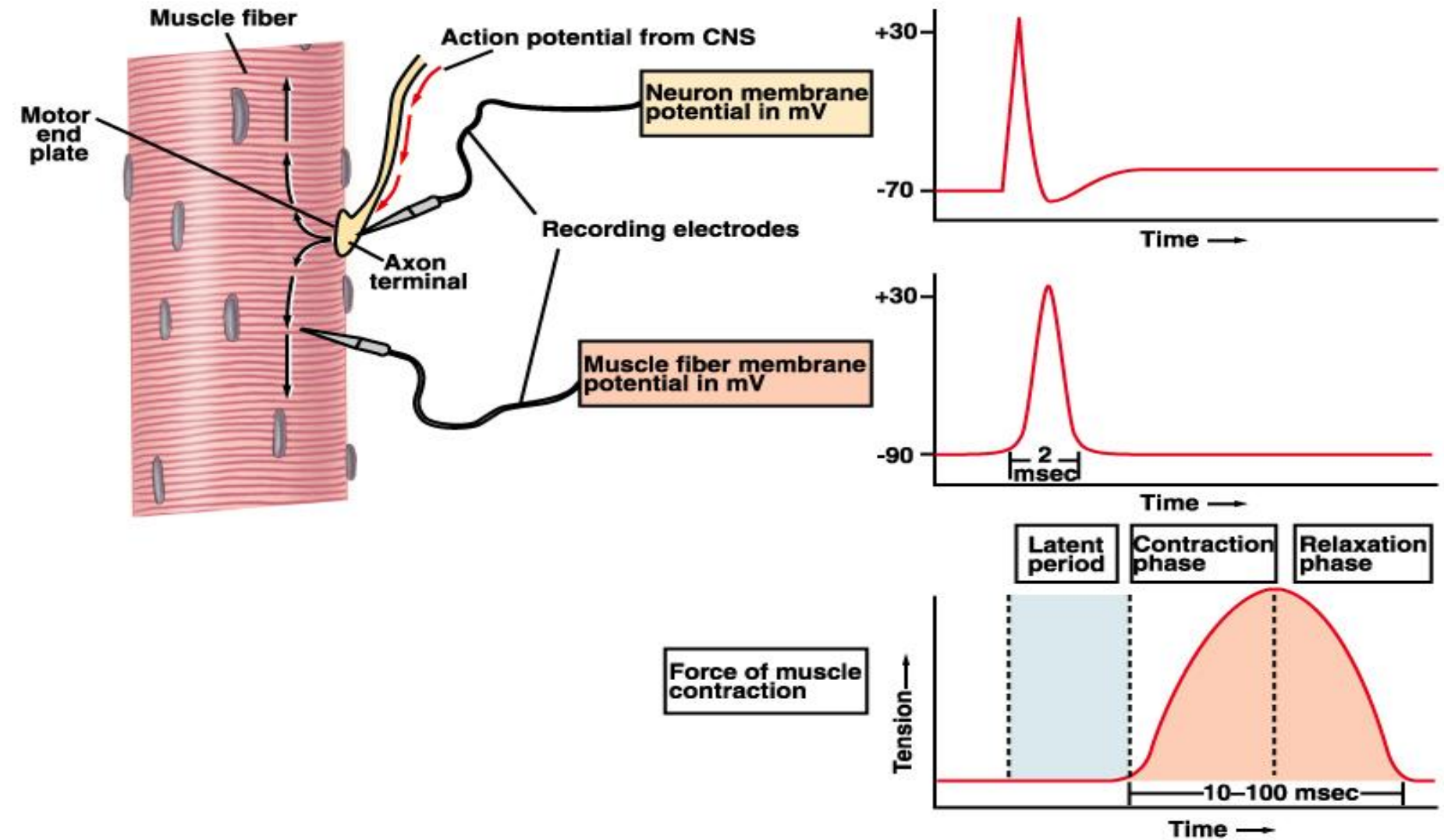
B. Contraction phase

- Beginning to end of muscle tension → myosin heads slide along the actin filaments

Muscle Contraction

C. Relaxation phase

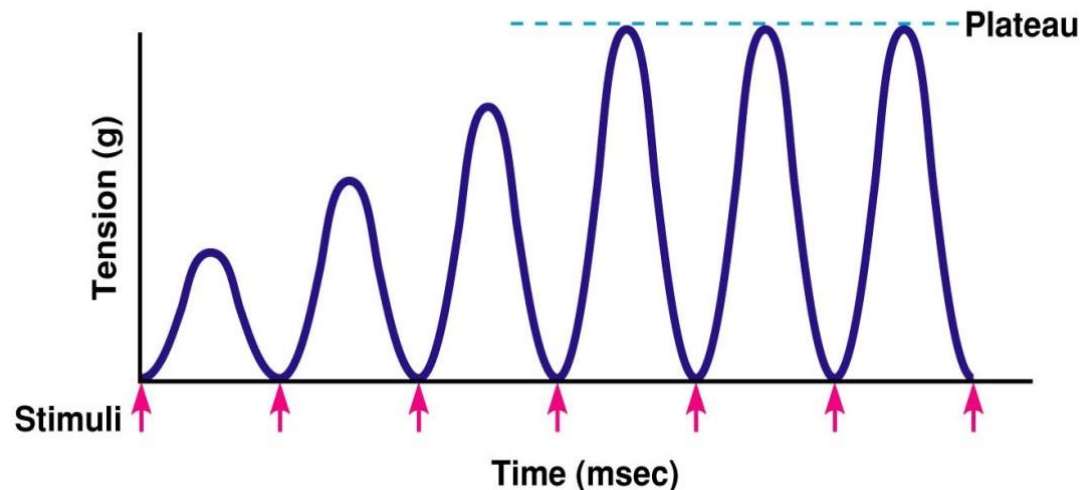
- From peak tension to no tension
- Ca^{2+} moved back into the cisternae
- Tropomyosin moves back over actin
- Myosin head release actin and the filaments move back into resting position



Muscle Contraction

2. Treppe

- Gradual increase in contraction intensity during sequential stimulation
- Occurs when successive **NON-OVERLAPPING** muscle twitches achieve greater force
- Due to Ca^{2+} accumulating in the cytoplasm with each stimulation

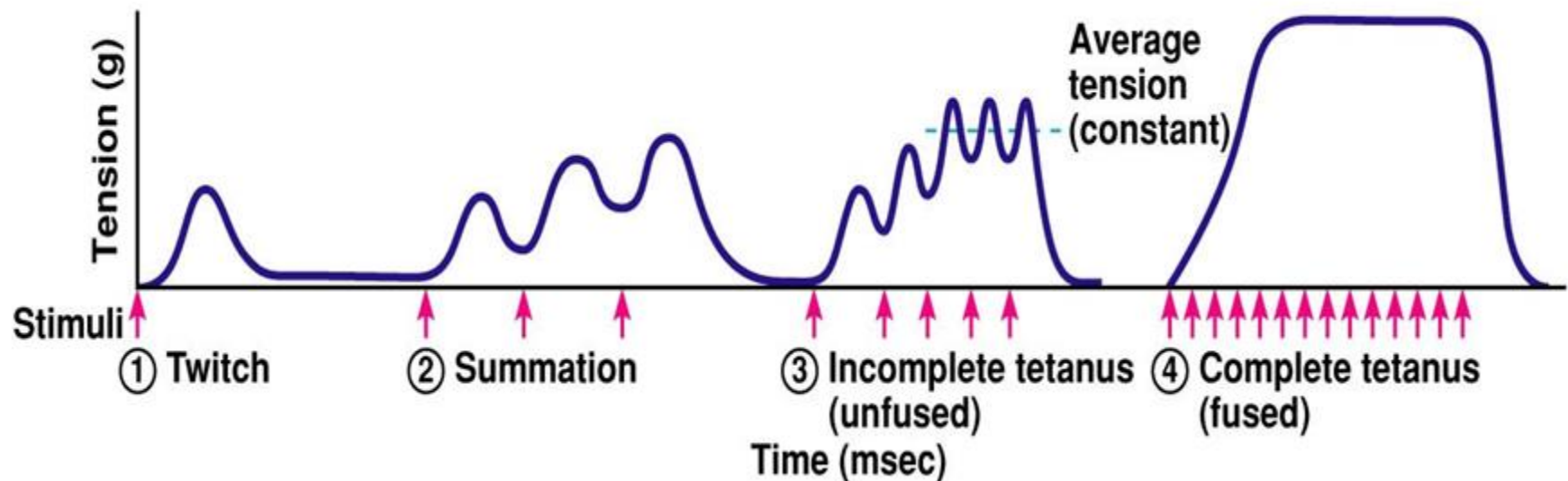


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Muscle Contraction

3. Summation

- Occurs when **OVERLAPPING muscle twitches** summate to produce a larger force
- Rapid sequence of stimuli → muscle twitches fuse into each other, each subsequent one being stronger than its precedent (due to Ca^{2+})



Muscle Contraction

4. Clonic contraction (Clonus, incomplete tetanus)

- If the stimuli are of moderate frequency just below the minimum tetanic frequency, and each stimulus falls in the relaxation phase, the result is a series of contraction with incomplete relaxation

5. Tetanic contraction (Tetanus)

- If high frequency stimuli are applied and fall on the contraction phase, the result is a continuous contraction without relaxation

Muscle Relaxation

- Breakdown of Ach by AchE
- Removal of Ca^{2+} from the Myoplasm ($<0.1\mu\text{mol/l}$) into the SR
 - The third ATP is consumed by Ca^{2+} -ATPase/**SERCA** for Ca^{2+} removal from the sarcoplasm
- After removal of Ca^{2+} :
 - Troponin returns to its original conformational state
 - Tropomyosin inhibition of Myosin-Actin interaction is restored
 - Cross-bridge cycling stops and the muscle is returned to its resting state

Steps in relaxation

Removal of Ach from nAChR



Release of Ca^{2+} from troponin



Ca^{2+} pumped back into SR



Cessation of interaction between actin and myosin



Relaxation

Clinical Correlates

Rigor Mortis

- **Rigor** → a state of muscle rigidity developed when muscle fibers are completely depleted of ATP and phosphorylcreatine
- **Rigor Mortis**
 - Several hours after death, all the muscles of the body goes into a state of contracture
 - The muscles contract and become rigid, even without APs
 - Ca^{2+} diffuses out of the SR & initiates muscle contraction using the remaining ATP molecules

Rigor Mortis

- This rigidity results from loss of all the ATP, which is required to recollect Ca^{2+} back into the SR and to disconnect myosin heads from actin, thus relaxation does not occur
- The muscles remain in rigor until the muscle proteins deteriorate about 15 to 25 hours later, which presumably results from autolysis caused by enzymes released from lysosomes

Muscular Atrophy

- Disuse atrophy
- Denervation atrophy
 - Paralysis (UMNL, LMNL)

Muscle Denervation

- Damage to spinal nerves produces **flaccid paralysis**, characterized by:
 - Reduced muscle tone
 - Depressed stretch reflexes and
 - Atrophy

Muscle Denervation

- Damage to UMNs or descending motor tracts at first produces **spinal shock** in which there is a **flaccid paralysis**
- When inhibition of UMNs on LMNs (α & γ) is removed, γ -motoneurons become hyperactive and the spindles become overly sensitive to stretch
- This is followed in a few weeks by **spastic paralysis**, characterized by:
 - Increased muscle tone
 - Exaggerated stretch reflexes + other signs of hyperactive LMNs
- The appearance of spastic paralysis suggests that UMNs normally exert an inhibitory effect on lower alpha and gamma motor neurons

Muscle Hypertrophy

- Increase in the size of muscle fiber
- Changes:
 - ↑No of actin and myosin filaments in each fiber
 - ↑No of sarcomeres
 - ↑Enzyme system of glycolysis
 - ↑Amount of myoglobin, ↑Mitochondria
 - ↑No of capillaries
- As a result, there is an increase in the force and strength of contraction, decrease in fatigability

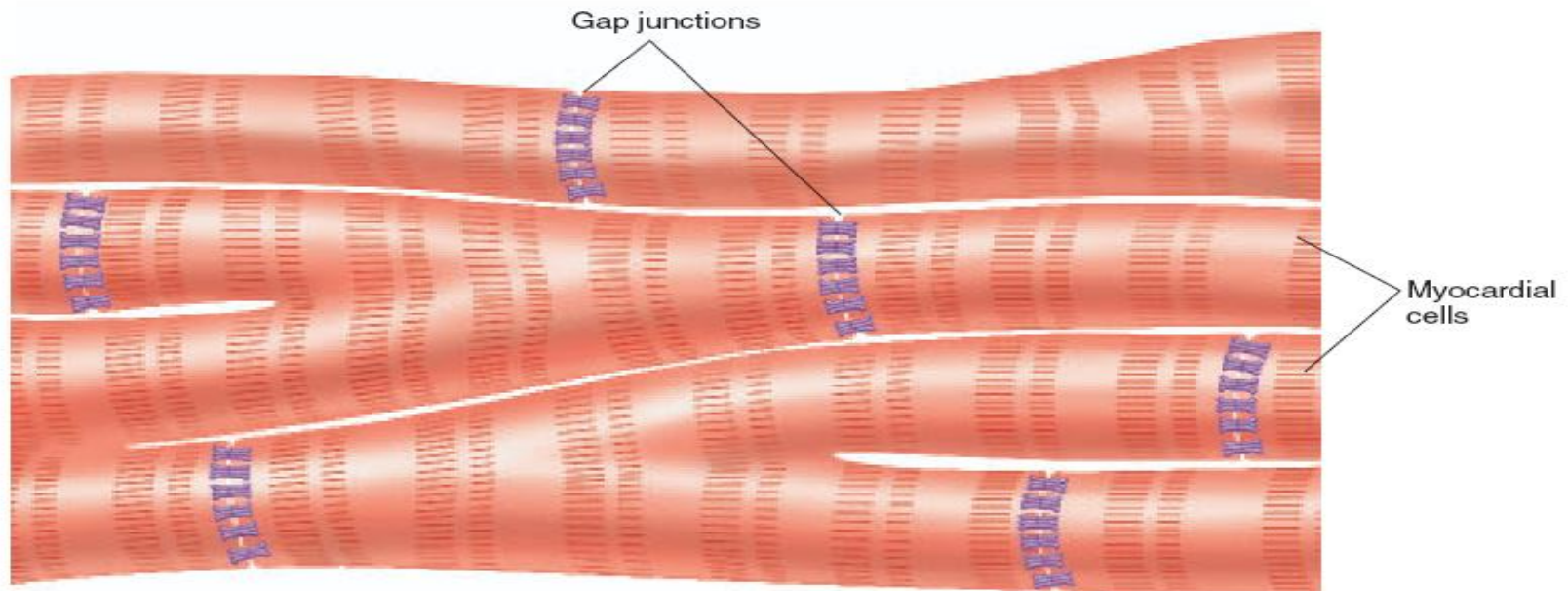
The Cardiac Muscle

Cardiac Muscle

- **Characteristics:**
 - Have cross striation (banding pattern of thick and thin filaments)
 - Abundance of connective tissue (prevent muscle rupture, prevent over-stretching during periods of increased venous return)
 - Has abundant amount of mitochondria + myoglobin which makes it **fatigue resistant**
 - **Auto-rhythmicity** and **syncytium** → have intercalated disc with gap junctions (syncytial bridges)
 - *Electrical + Mechanical coupling*

Cardiac Muscle

- Characteristics:
 - In the course of an average life span the heart will contract = 72 beats $\times$ 60 min $\times$ 24hrs $\times$ 365d $\times$ 70 years = 3×10^9
 - Involuntary muscle tissue (ANS + Hormones...)



Myocardial Innervation

1. Sympathetic

- Innervate both atrium & ventricles
- NE binds to adrenoceptors in the heart (*affinity: $\beta_1 > \beta_2 & \alpha_1$*)
- Sympathetic stimulation of the myocardium produce **Positive Inotropy** ($\uparrow$ force of contraction)

Myocardial Innervation

2. Parasympathetic Innervation

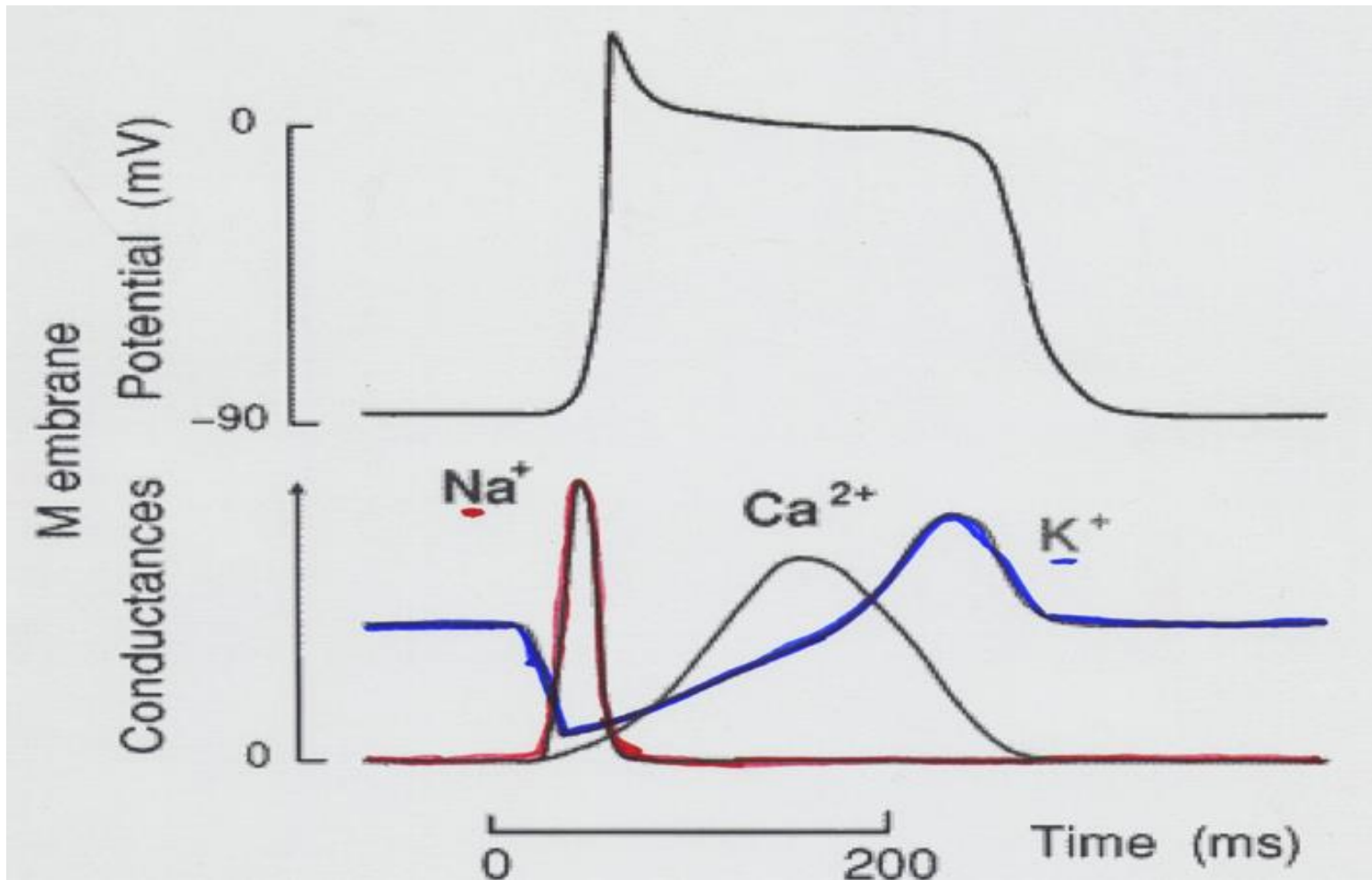
- Innervate atrium (lesser extent) + ventricular muscle (very little)
- Ach binds to postsynaptic muscarinic receptors (M_2) in the heart and produce **negative Inotropy**
- Strong stimulation of vagus, completely blocks excitatory signal transmission to the ventricles
- After a moment, however, purkinje fibers develops a rhythm of its own and causes ventricular contraction (*Ventricular escape*)

Excitation-Contraction Coupling

- **Excitation:**
 - Spread of excitation from cell-to cell via gap junctions + spread to the interior via T-tubules
 - During plateau phase, Ca^{++} permeability increases $\rightarrow$ this Ca^{++} triggers release of Ca^{++} from the SR $\rightarrow$ $\uparrow\text{Ca}^{++}$ level in the cytosol

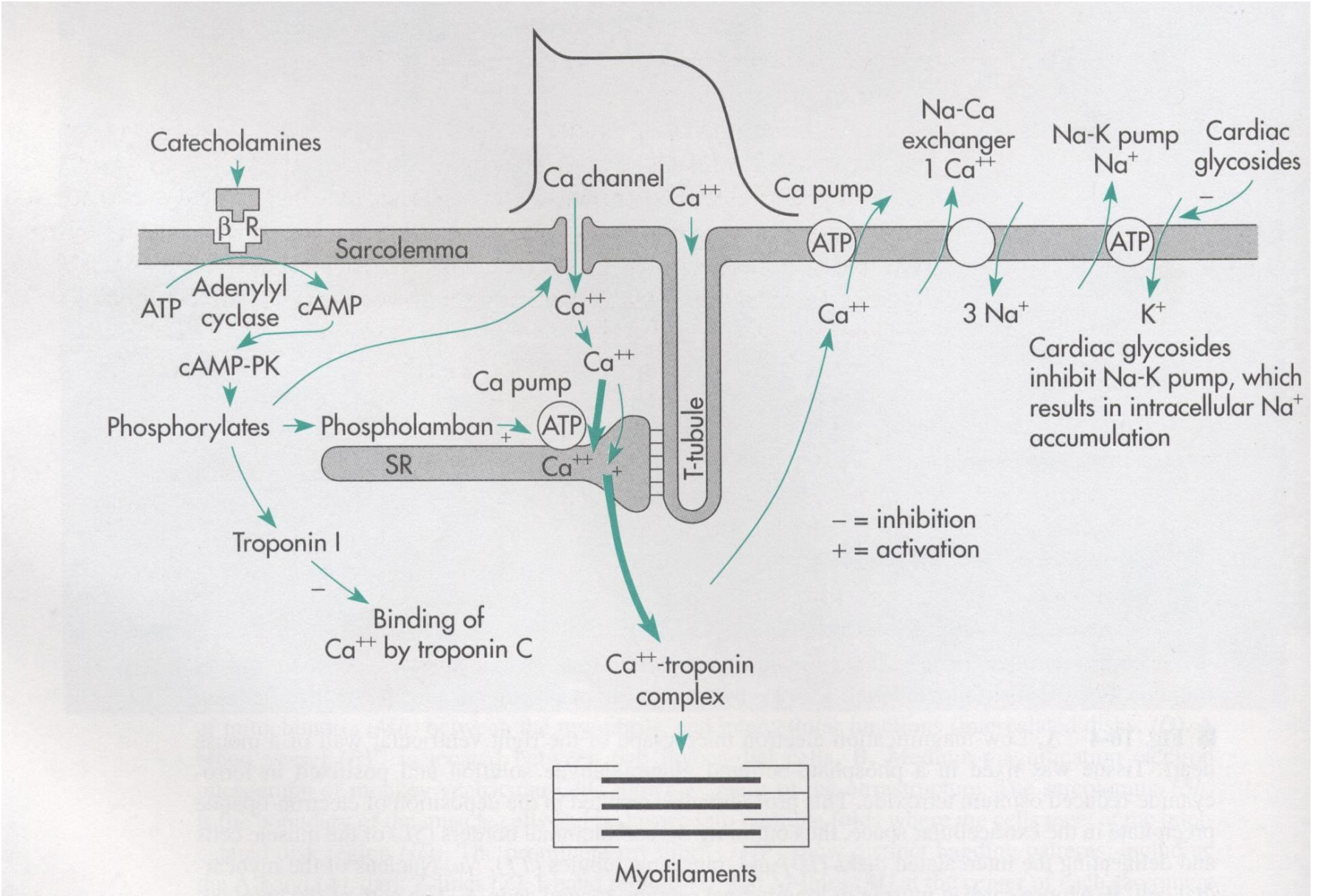
Excitation-Contraction Coupling

- Excitation:



Excitation-Contraction Coupling

- Release of Ca^{2+}
 - Ca^{2+} - release from the SR is triggered by Ca^{2+} (Ca^{2+} induced release Ca^{2+})
 - T-Tubule (DHPR) contains Ca^{2+} channel (through which Ca^{2+} enters the cell during stimulation by the AP)
 - SR-RyR containing Ca^{2+} - release channel is opened by Ca^{2+} influx from the T-Tubule DHPR



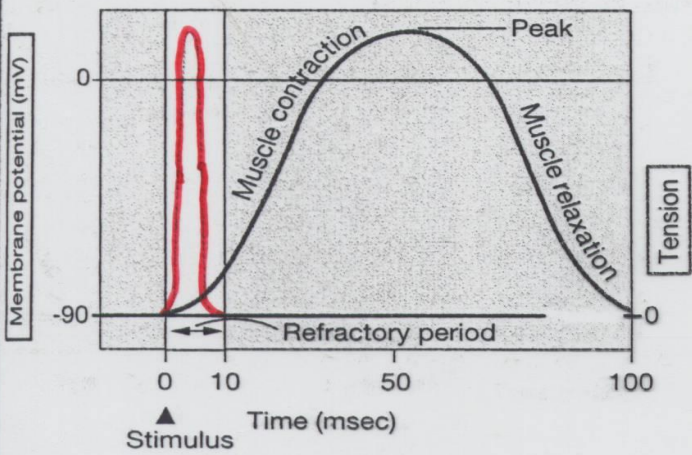
Contraction (Systole)

- Ca^{++} binds to Troponin C
- Ca^{++} -Troponin complex interacts with tropomyosin to unlock active site between actin and myosin
- Cross bridge cycling → contraction (systole)

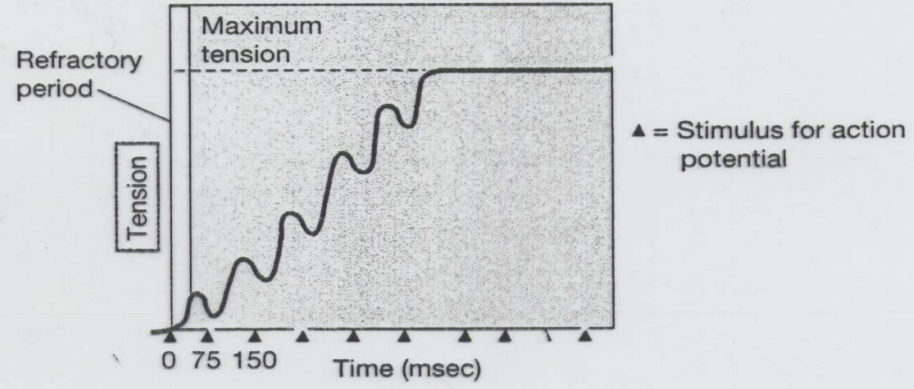
Contraction (Systole)

- Nature of muscle contraction:
 - Skeletal muscle
 - Tetanic contraction possible
 - Short refractory period
 - Cardiac muscle
 - No tetanic contraction
 - Long refractory period

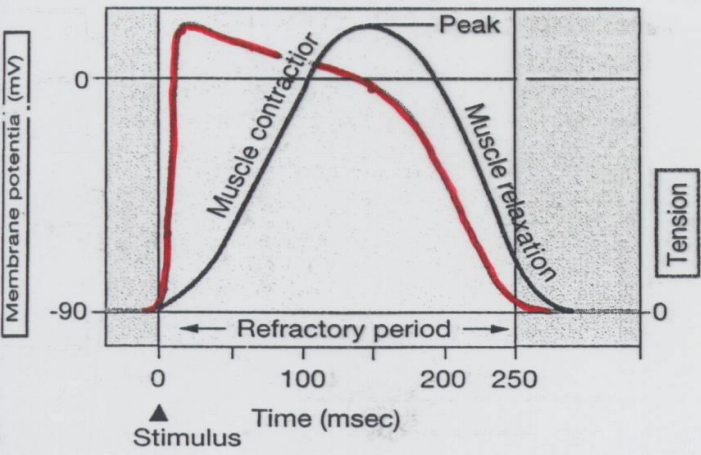
(a) Skeletal muscle fast-twitch fiber



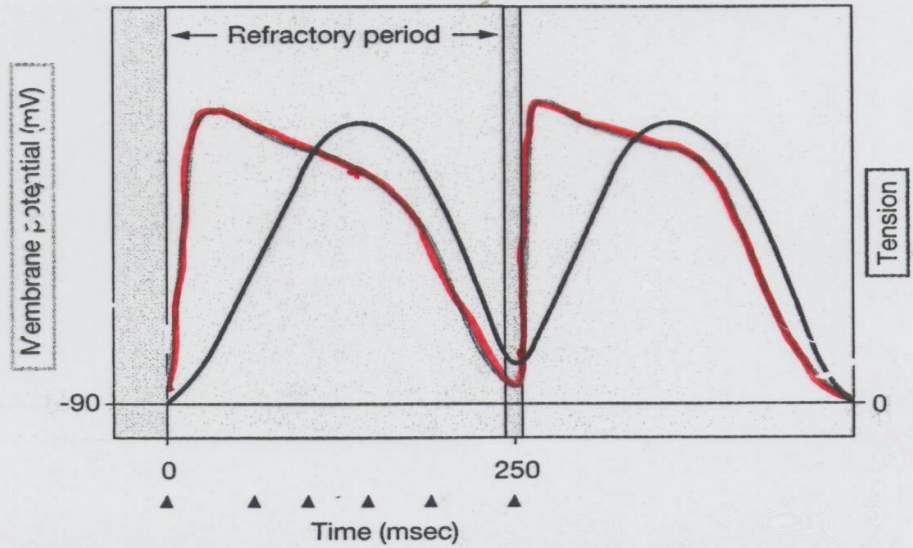
(b) Tetanus in a skeletal muscle. Action potentials not shown.



(c) Cardiac muscle fiber

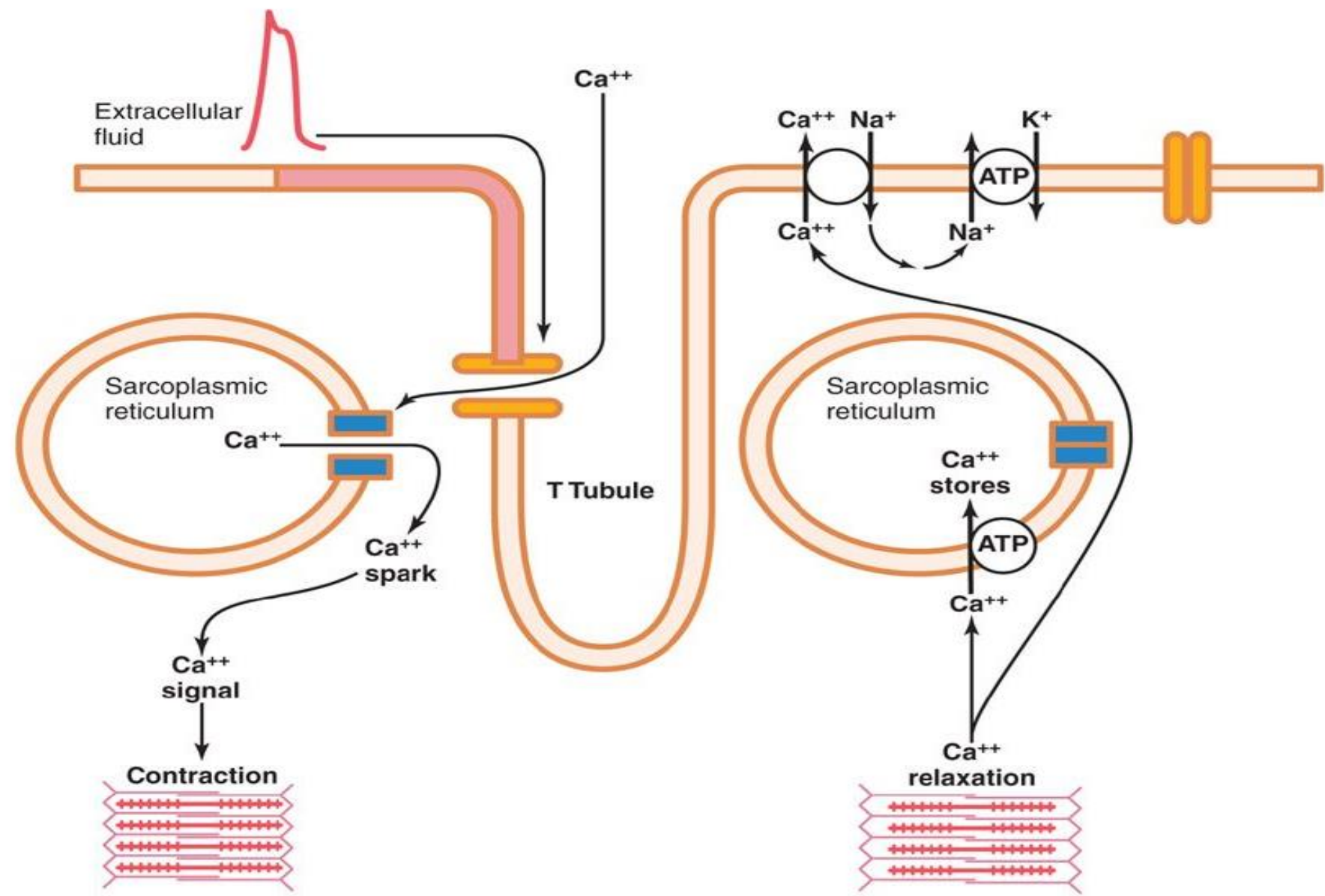


(d) Long refractory period in a cardiac muscle prevents tetanus.



Relaxation (Diastole)

- As a result of Ca^{++} removal from the cytosol by:
 - Uptake to SR by SERCA (inhibited by Phospholamban)
 - Extrusion by Na^+ - Ca^{++} exchange
 - Ca^{++} pump (to limited extent)
- Hormonal: inhibition of Troponin-C- Ca^{++} bondage by Troponin-I



Effects of non-physiological environment:

1. pH:

- **Acidosis**

- - ve Inotropic effect, due to ↓affinity of troponin C to Ca^{++}
- Severe acidosis stops heart in diastole

- **Alkalosis**

- +ve inotropic effect due to ↑ affinity of troponin C to Ca^{++}
- Severe alkalosis stops heart in systole

Cardiac Muscle

2. Na^+

- **Hypernatremia**
 - -ve inotropic effect
 - Stimulate Na^+ - Ca^{++} exchanger to take Na^+ in, drive Ca^{++} out of cardiac myocyte $\rightarrow$ cytosolic Ca^{++} level decreases
- **Hyponatremia** $\rightarrow$ opposite effect

Cardiac Muscle

3. Ca^{2+}

- **Hypercalcemia:**
 - Increase cytosolic Ca^{++} level, stronger systole & incomplete diastole
 - If the level is very high, heart stop in systole (Calcium Rigor)
- **Hypocalcemia:**
 - Decrease myocardial contractility, but have no serious effect

Cardiac Muscle

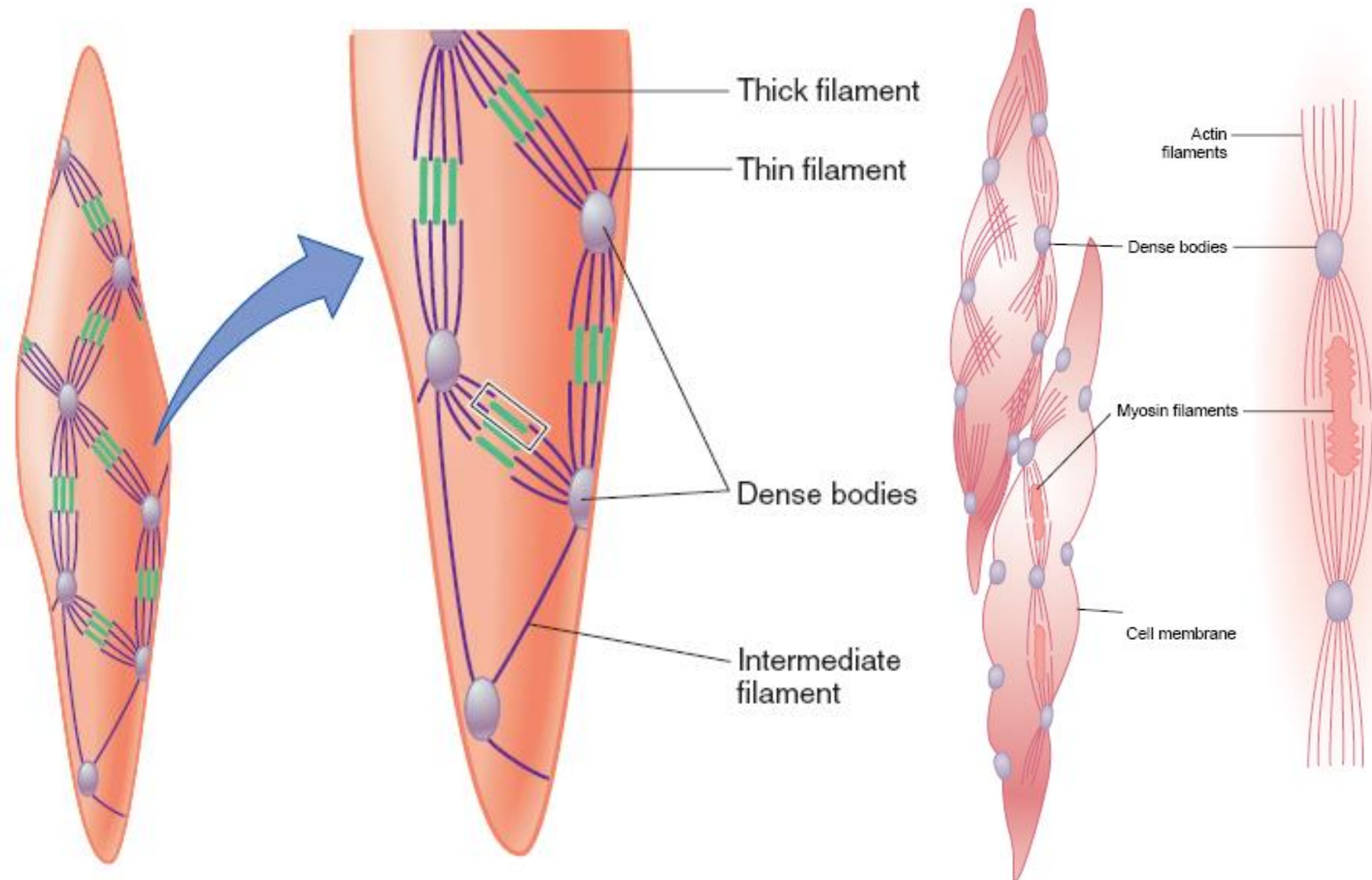
4. K^+

- **Hyperkalemia:**
 - -ve inotropic effect
 - Low amplitude AP $\rightarrow$ weak contraction
 - Marked hyperkalemia heart stop in diastole
- **Hypokalemia** $\rightarrow$ weak + inotropic effect

Smooth Muscle

Smooth Muscle

- **Characteristics:**
 - It has no striations (sarcomeres are absent)
 - Thick filaments → myosin
 - Thin filaments → actin and tropomyosin (No troponin)
 - Thin filaments are attached to dense bodies (*functional equivalents of Z-disc*)
 - Dense bodies are found in the cell membrane and cytoplasm composed of actinin
 - T-tubules are absent



Smooth Muscle

- **Characteristics:**
 - Involuntary muscle tissue (ANS + Hormones...)
 - It is important in regulation of the airways, blood vessels, GIT, and hollow organs (bladder, uterus...)
 - It produces **slower and longer-lasting contractions:**
 - ↓rates of cross-bridge cycling → LATCH STATE → maintain TONE and little energy consumed

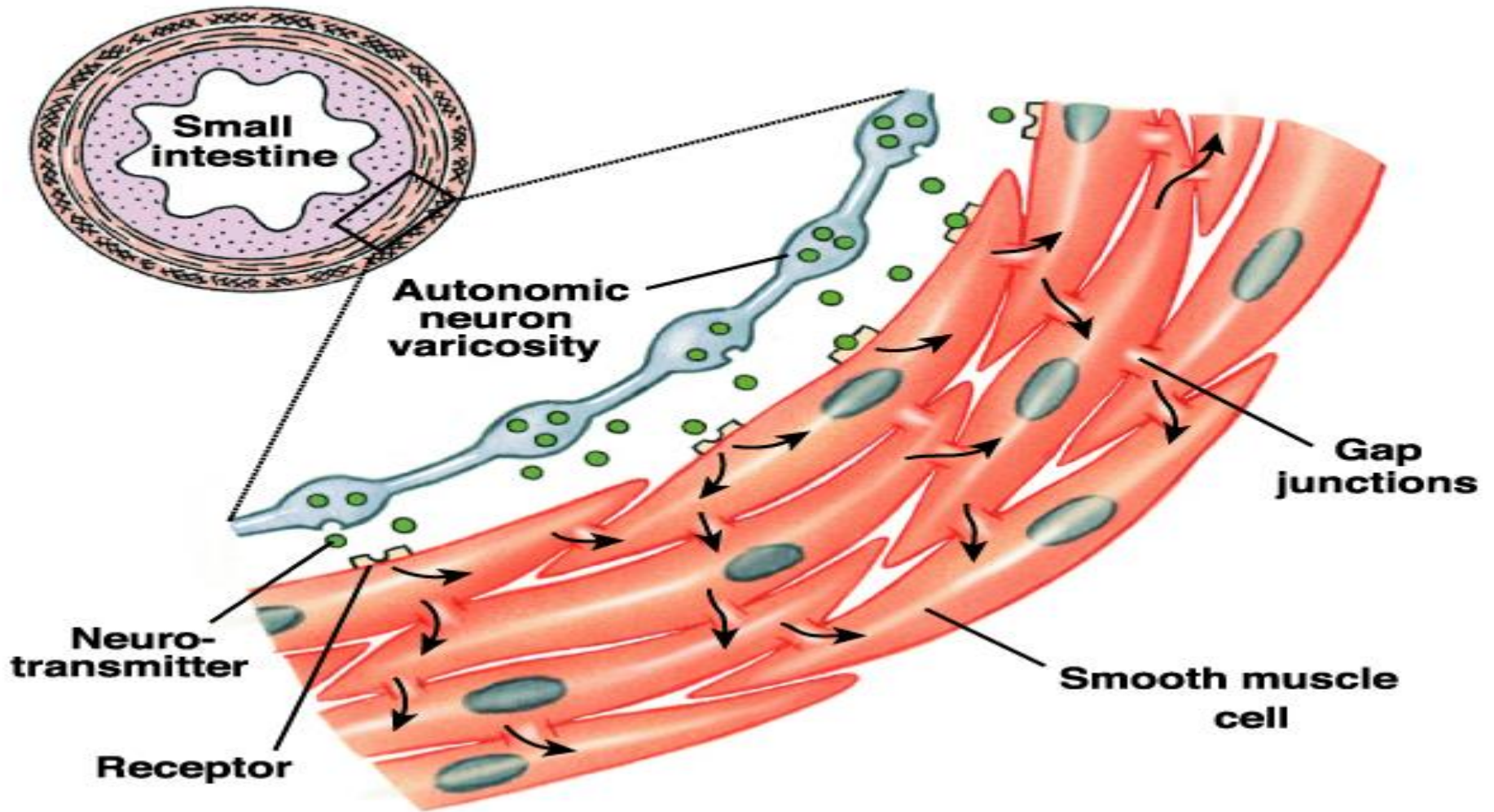
Smooth Muscle

- Classification:

1. Single-unit smooth muscle (Visceral smooth muscle)

- A mass of hundreds to thousands of smooth muscle fibers arranged in sheets or bundles that contract together as a single unit
- Connected to one another by low resistance gap junctions:
 - Synchronous excitation and contractions (*functional syncytium*)
- Some contract rhythmically due to pacemaker cells
- Found primarily in the walls of hollow viscera (gut, ureter, uterus, and small blood vessels...)

(a) Single-unit smooth muscle

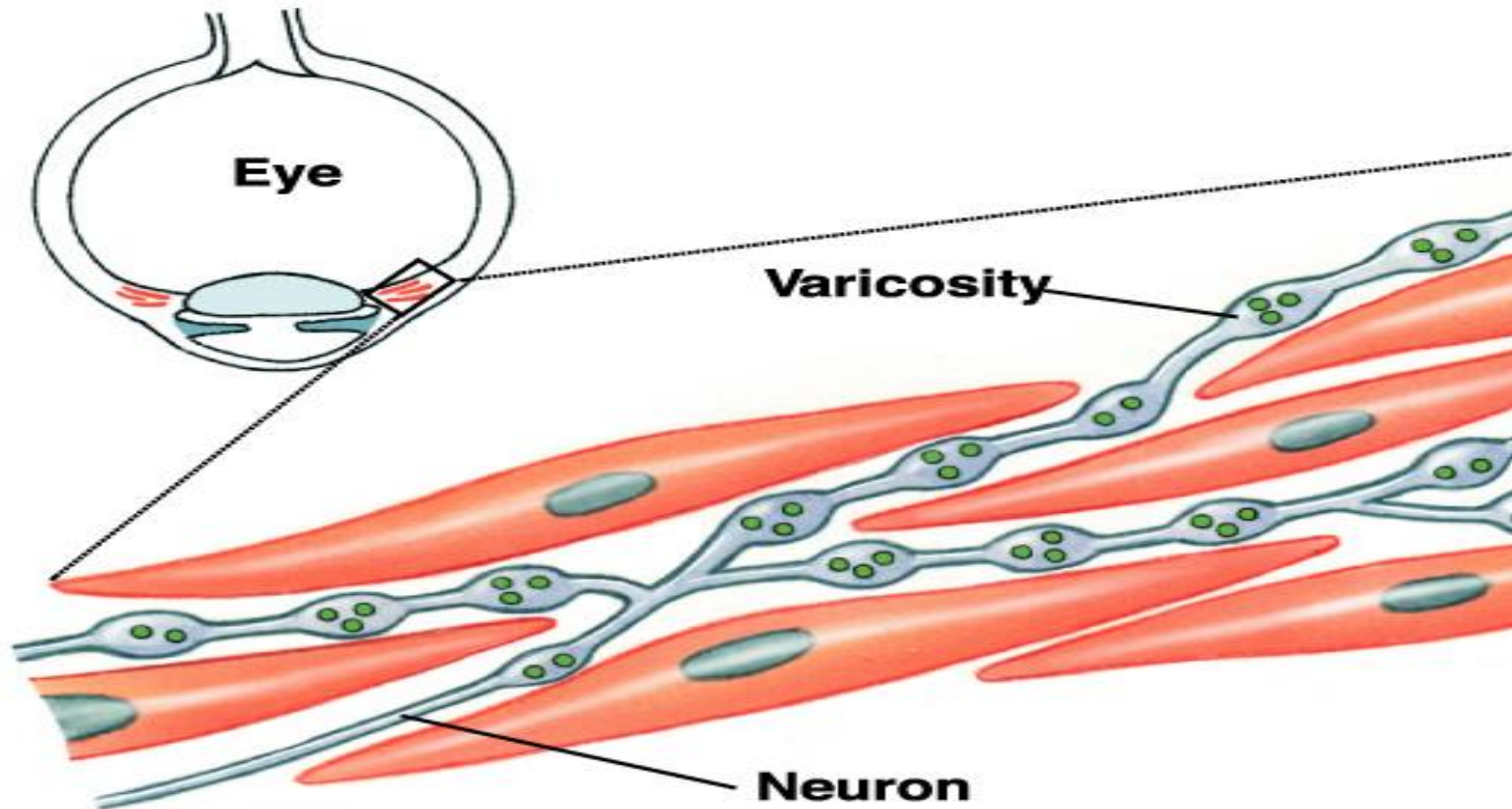


Smooth Muscle

2. Multi-unit smooth muscle

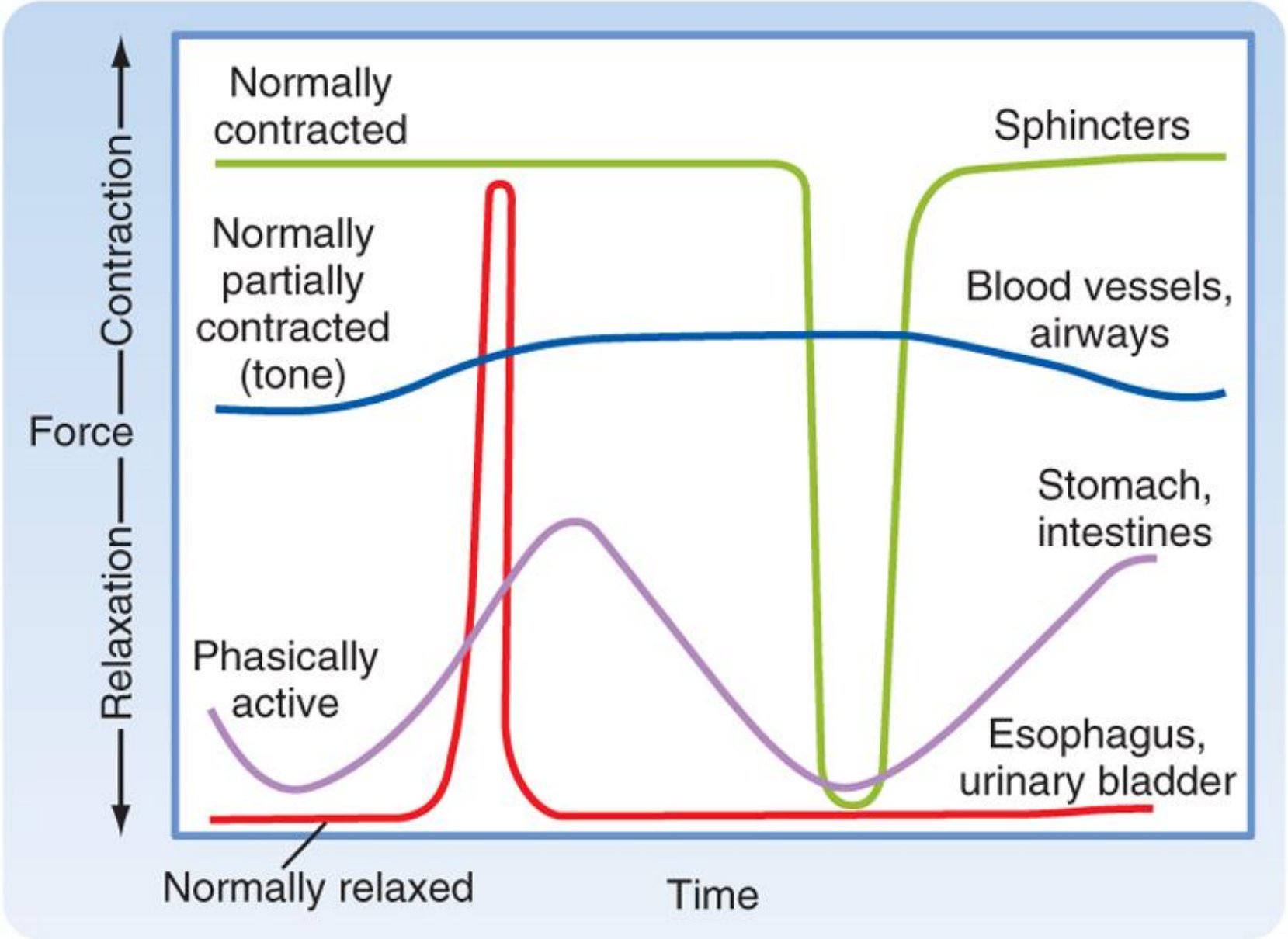
- Composed of discrete, separate smooth muscle fibers
- Each fiber is independent of all the others (*have no gap junctions*)
- Each fiber receives a **single nerve ending**, like skeletal muscles
- **No pacemaker cells**
- Found in **Ciliary and iris muscles of the eye, Piloerector, large blood vessels**

(b) Multi-unit smooth muscle



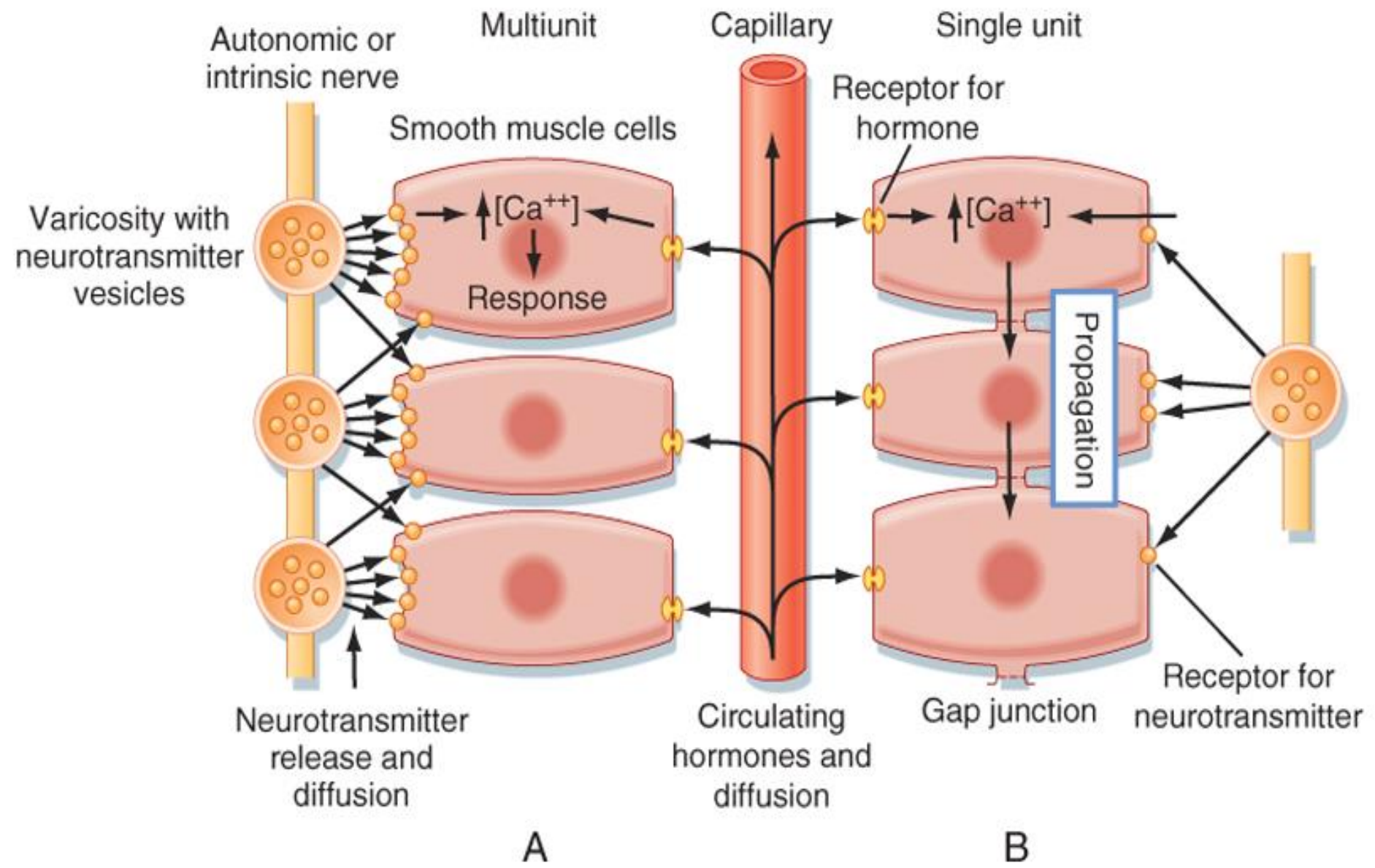
Smooth Muscle

- Smooth muscle types based on **muscle tone**:
 1. Normally contracted
 2. Normally partially contracted
 3. Phasically active
 4. Normally relaxed



Neural Control of Smooth Muscle

- Smooth muscle membrane contains many types of receptors that can be stimulated/inhibited by multiple types of signals:
 - Hormonal stimulation
 - Stretch of the muscle...
 - Neural stimulation (ANS)
 - Autonomic nerve fibers that innervate smooth muscles generally branch diffusely on top of a sheet of muscle fibers
 - NTs secreted by the autonomic nerves innervating smooth muscles are **Ach & NE** (usually function **antagonistically**)

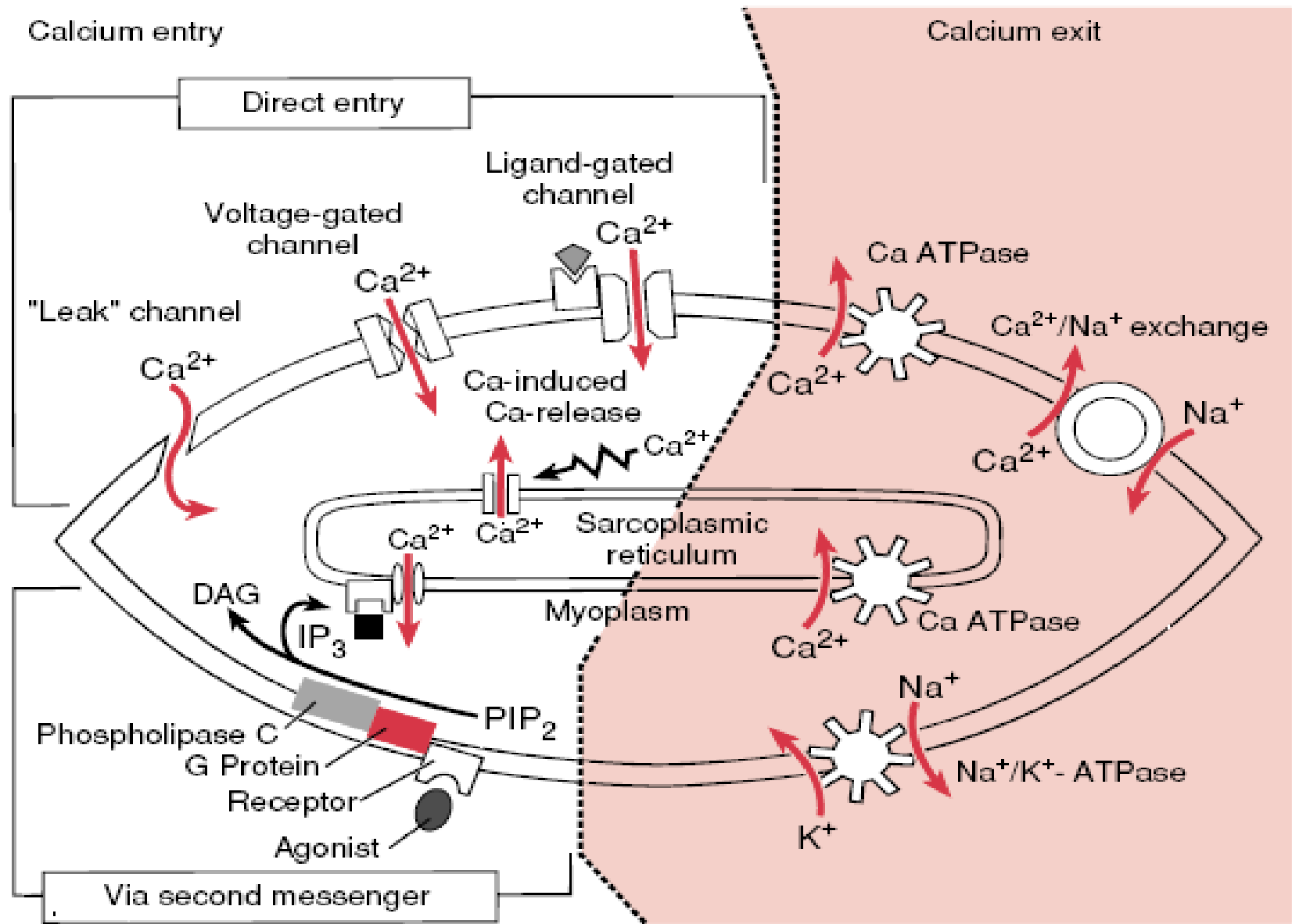


Molecular Basis of Contraction

- Calcium Channels
 - Smooth muscles have far more voltage-gated Ca^{2+} channels but few voltage-gated Na^{+} than skeletal muscles
 - Thus, Na^{+} has little effect in the generation of AP in most smooth muscles
 - Ca^{2+} channels open more slowly than Na^{+} channels and remain open much longer (accounts for the prolonged plateau)
 - Ca^{2+} entry into the cell during the AP also acts directly on the smooth muscle contractile machinery to cause contraction

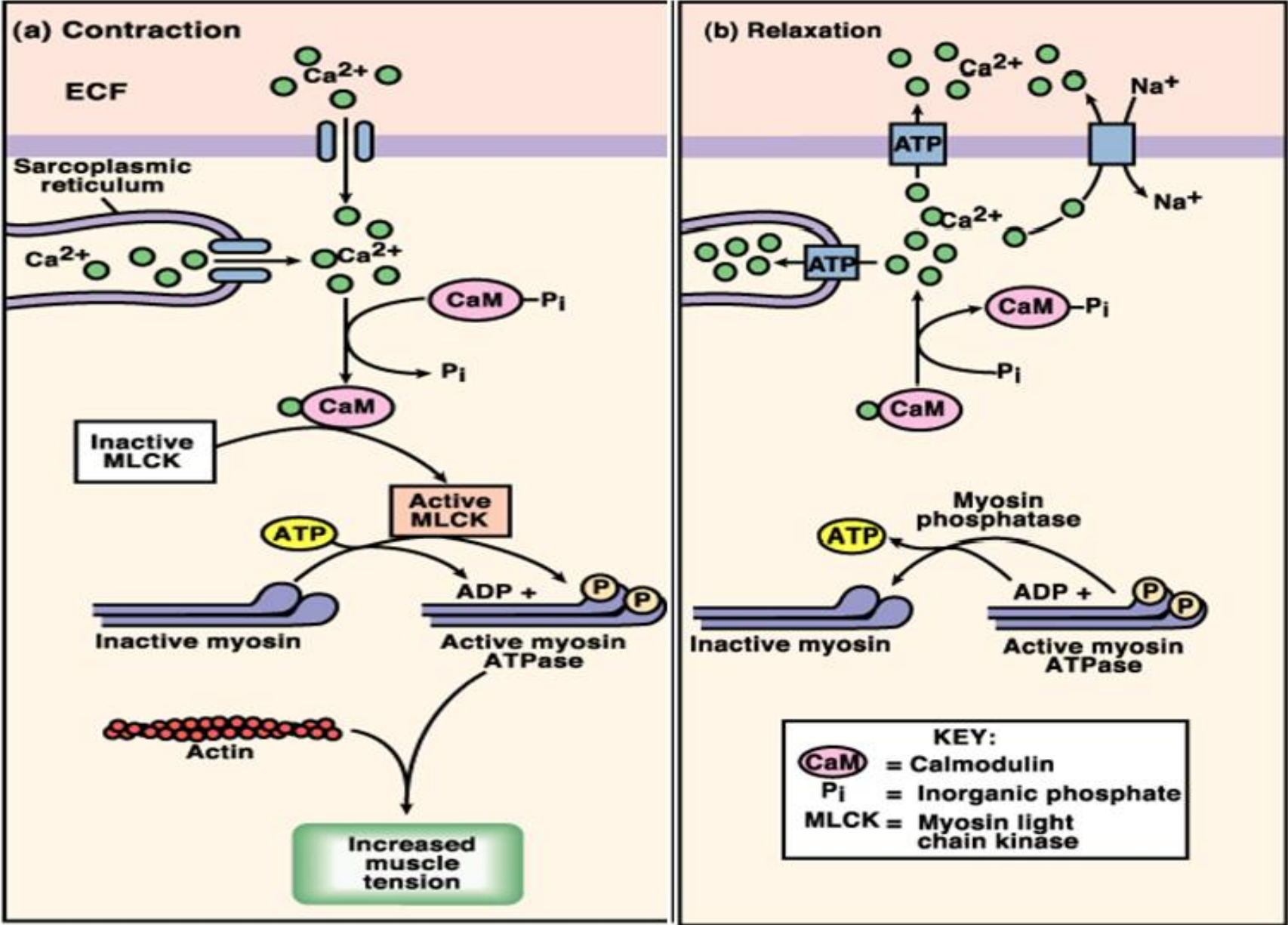
Molecular Basis of Contraction

- Ca^{2+} can enter the cell in a variety of ways:
 - Ligand gated Ca^{2+} channels
 - Voltage gated Ca^{2+} channels
 - Leak channels
 - Release from the SR by RyR + IP_3R



Molecular Basis of Contraction

- Activation of muscle filaments:
 - Cross-bridge cycling is regulated by Ca^{2+} induced phosphorylation of myosin
 - Myosin cross-bridge has 4 light chains
 - Myosin can not bind to actin unless one of these light chains (LC_{20}) is phosphorylated
 - LC_{20} is Phosphorylated by Myosin Light Chain Kinase (MLCK)
 - MLCK is activated by Ca^{2+} -Calmodulin complex
 - Phosphorylation of LC_{20} enables the myosin to interact with actin
 - LC_{20} is dephosphorylated by MLCP



Molecular Basis of Relaxation

- When the Ca^{2+} concentration falls below a critical level, the contraction processes automatically reverses, except for the phosphorylation of the myosin head
- **Myosin phosphatase** dephosphorylates the regulatory light chain and stops the cycling and contraction
- The time required for relaxation of muscle is determined to a great extent by the amount of active myosin phosphatase in the cell
- Calcium pumped out of the cell $\rightarrow$ $\downarrow$ active calmodulin $\rightarrow$ $\downarrow$ active MLCK $\rightarrow$ $\downarrow$ number of cross-bridges $\rightarrow$ relaxation

